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(54) SEMICONDUCTOR DEVICE INCLUDING ACTIVE PATTERN AND CONDUCTIVE PAD PATTERN

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(52) **U.S. Cl.**
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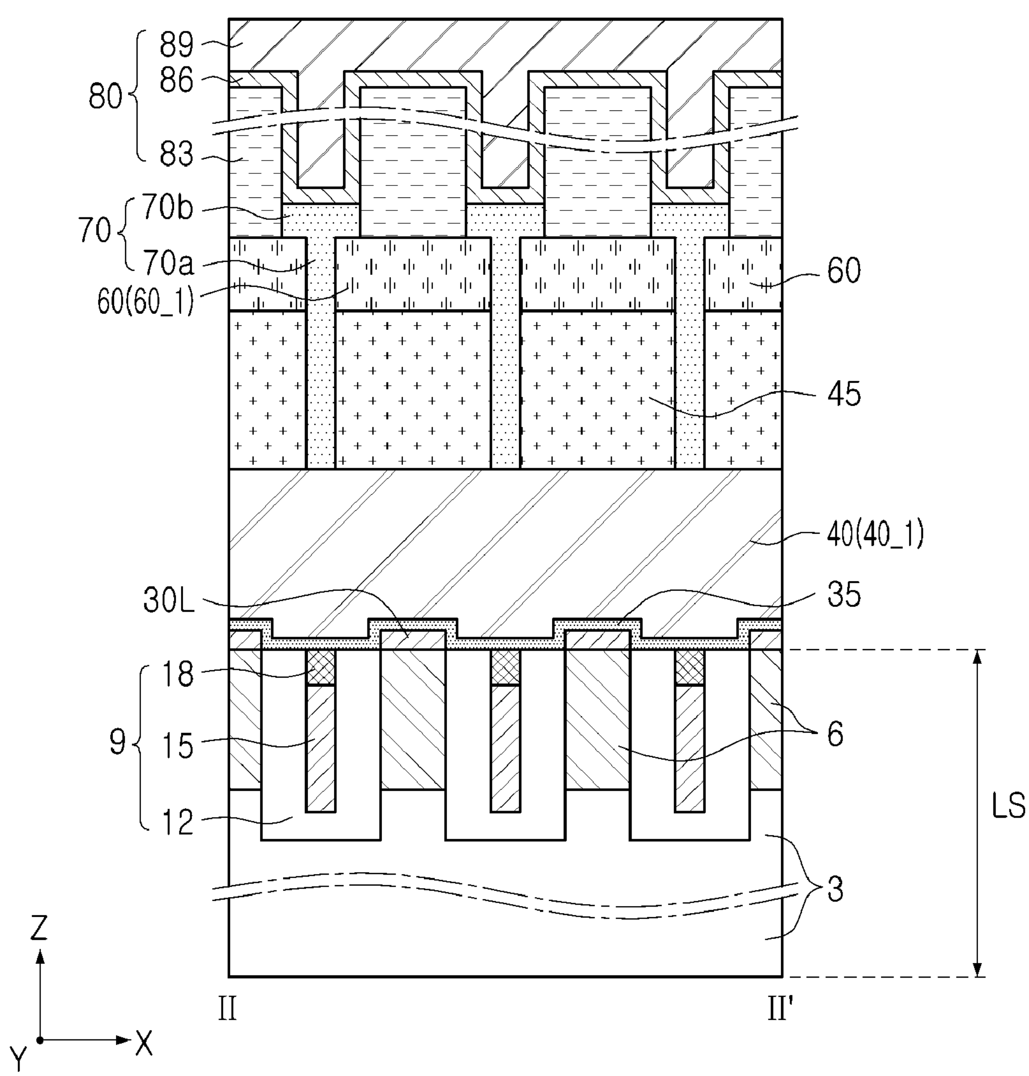
(57) **ABSTRACT**

A semiconductor device includes a bit line, an active pattern on the bit line, and including a lower active portion electrically connected to the bit line, a first vertical active portion extending upwardly from the lower active portion, and a second vertical active portion extending upwardly from the lower active portion, a word line between a lower region of the first vertical active portion and a lower region of the second vertical active portion, and a conductive pad pattern electrically connected to an upper region of the first vertical active portion and an upper region of the second vertical active portion.

1



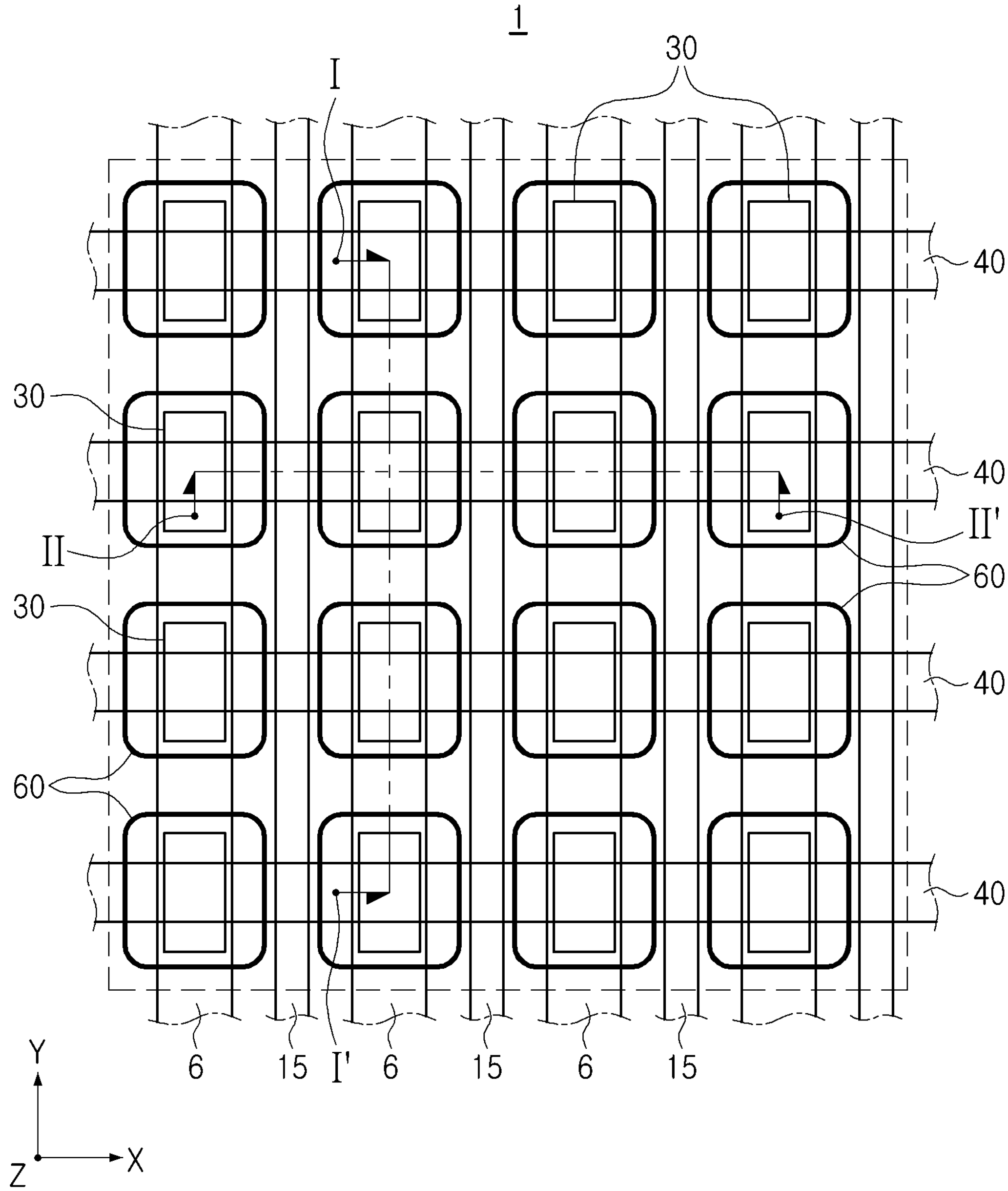
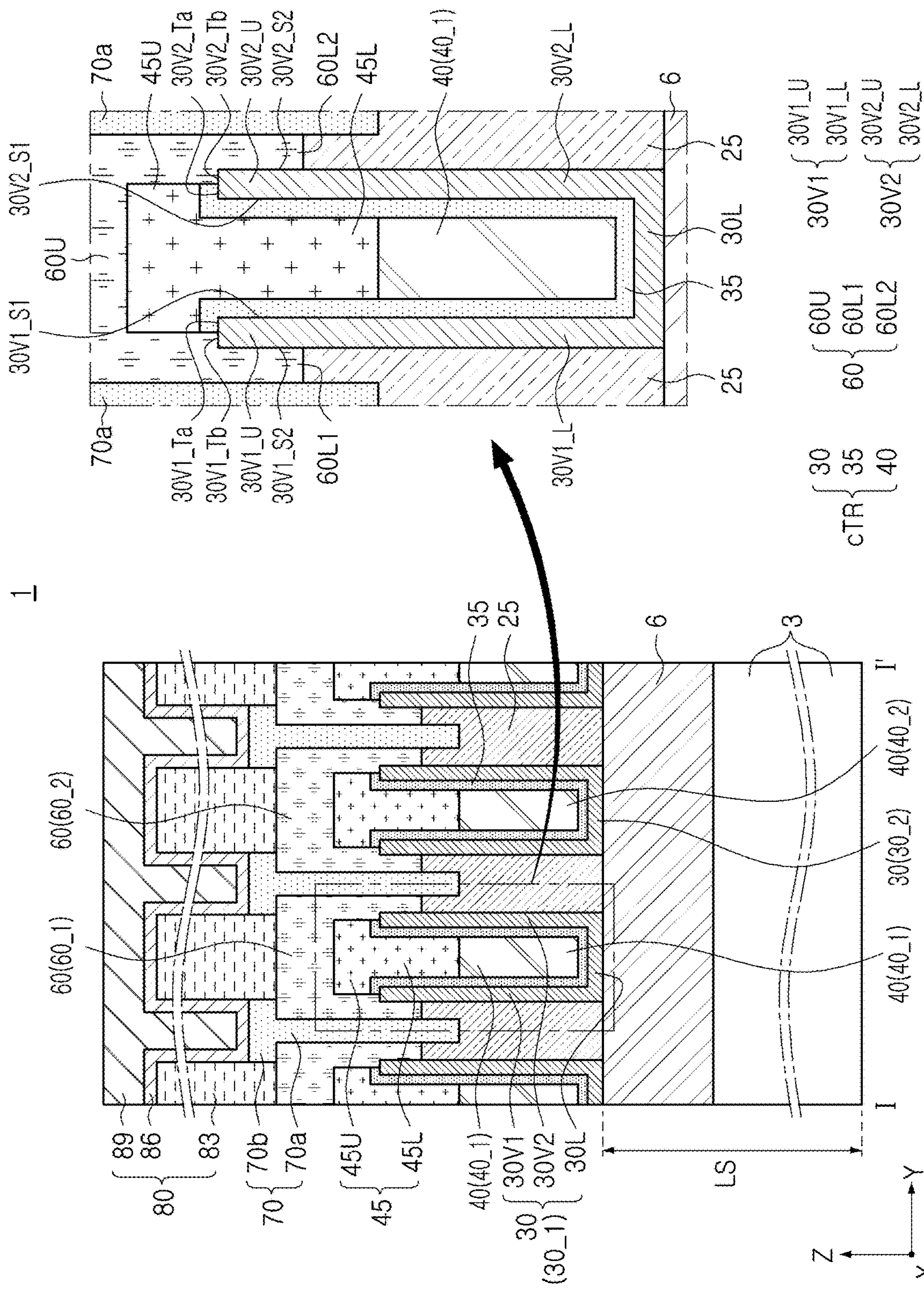



FIG. 1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89
86
83
80
70b
70a
70
45U
45L
45
40(40_1)
30V1
30V2
30L
30
(30_1)
LS
60(60_1)
60(60_2)
35
25
6
3
40(40_1)
30(30_2)
40(40_2)
I
I'
Z
Y
X
70a
45U
30V2_Ta
30V2_Tb
30V2_U
30V2_S2
60L2
45L
40(40_1)
30V2_L
6
30V2_S1
60U
30V1_S1
25
30L
35
30V1_Ta
30V1_Tb
30V1_U
30V1_S2
60L1
30V1_L
30V1 { 30V1_U 30V1_L
30V2 { 30V2_U 30V2_L
60 { 60U 60L1 60L2
cTR { 30 35 40

FIG. 2A

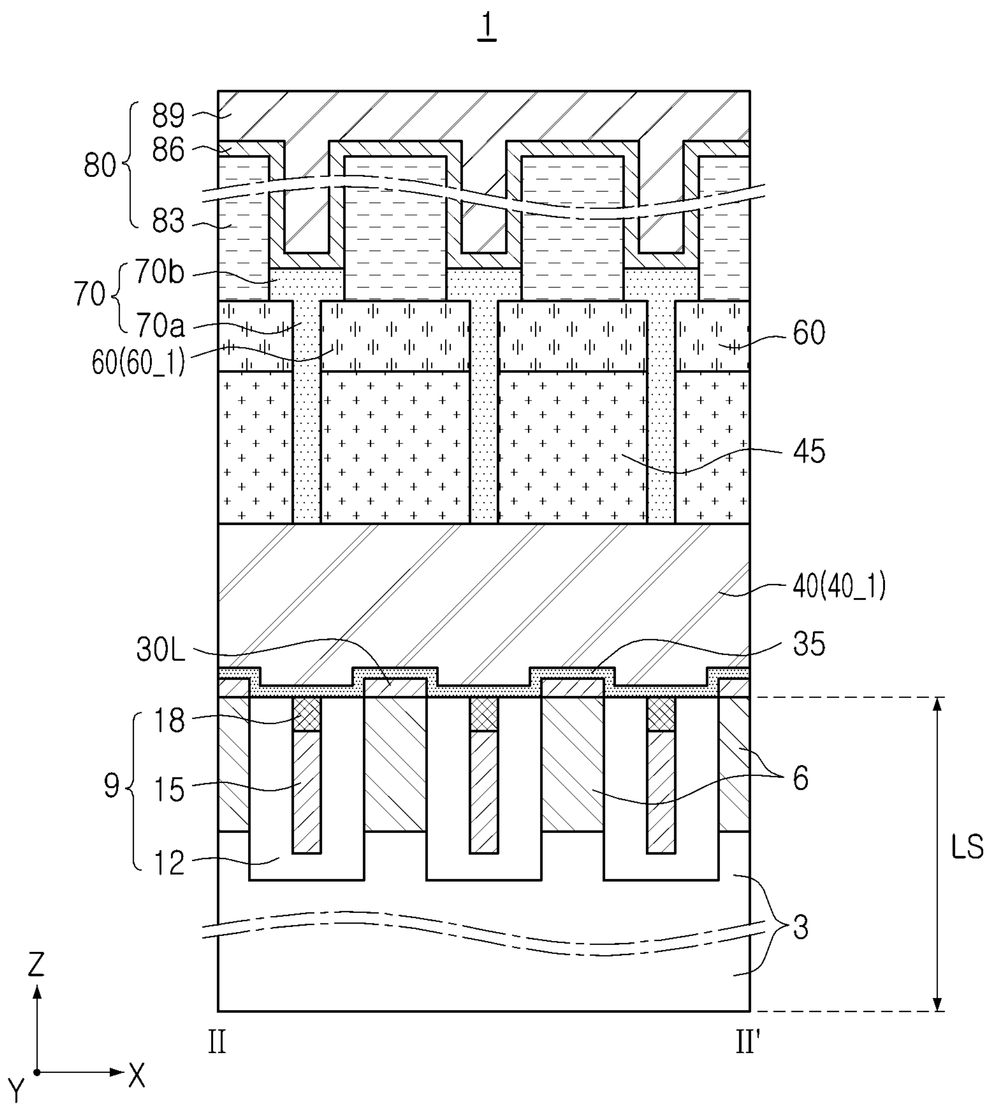

1
89
86
80
83
70b
70
70a
60(60_1)
60
45
40(40_1)
30L
35
18
9
15
6
12
LS
3
Z
Y
X
II
II'

FIG. 2B

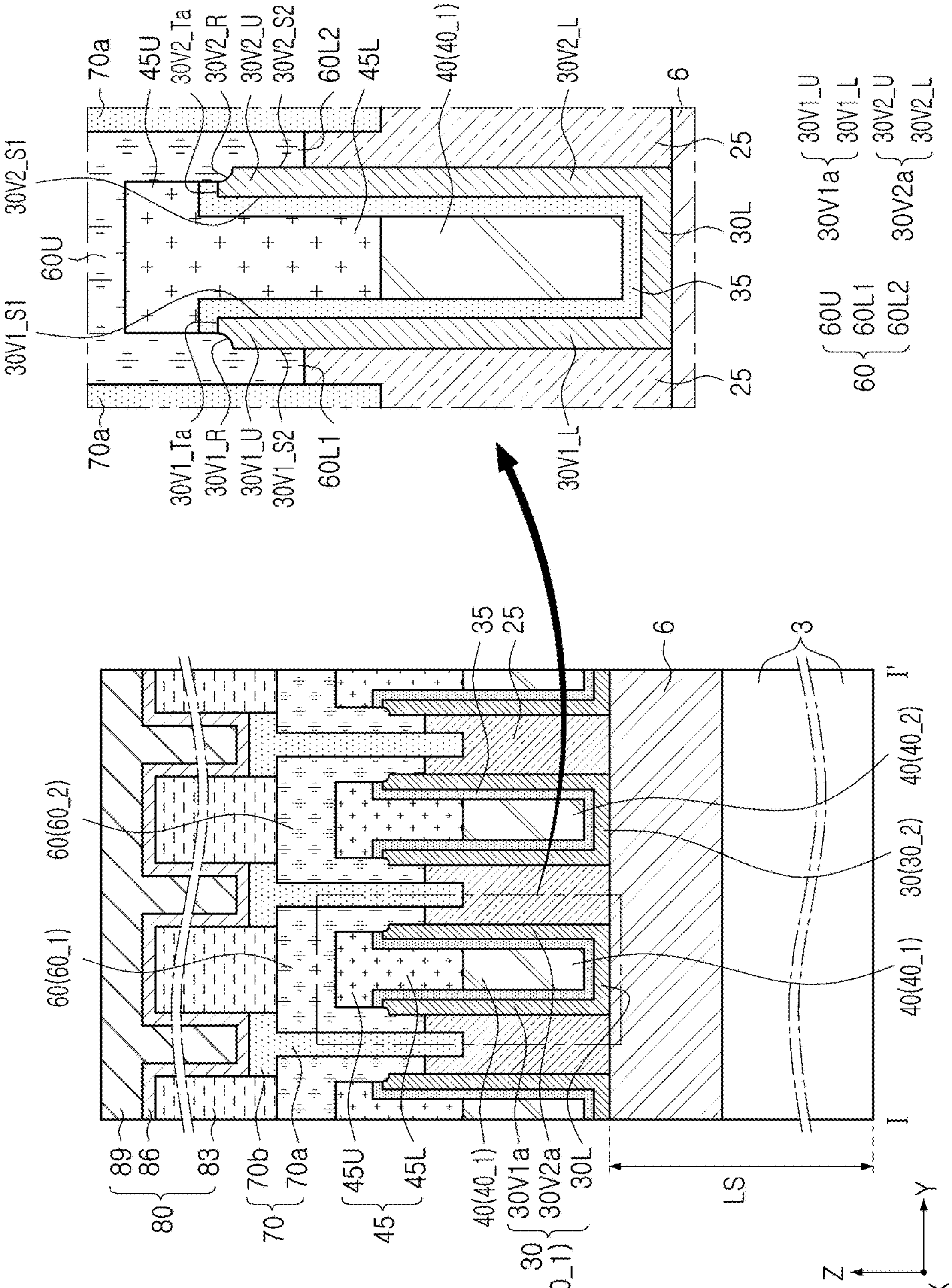

70a
45U
30V2_Ta
30V2_R
30V2_U
30V2_S2
60L2
45L
40(40_1)
30V2_L
6
30V2_S1
60U
30V1_S1
25
30L
35
30V1_Ta
30V1_R
30V1_U
30V1_S2
60L1
30V1_L
30V1a
30V1_U
30V1_L
30V2a
30V2_U
30V2_L
60
60U
60L1
60L2
60(60_1)
60(60_2)
3
I
I'
89
86
83
80
70b
70
45
30
(30_1)
LS
40(40_2)
30(30_2)
Z
Y
X

FIG. 3

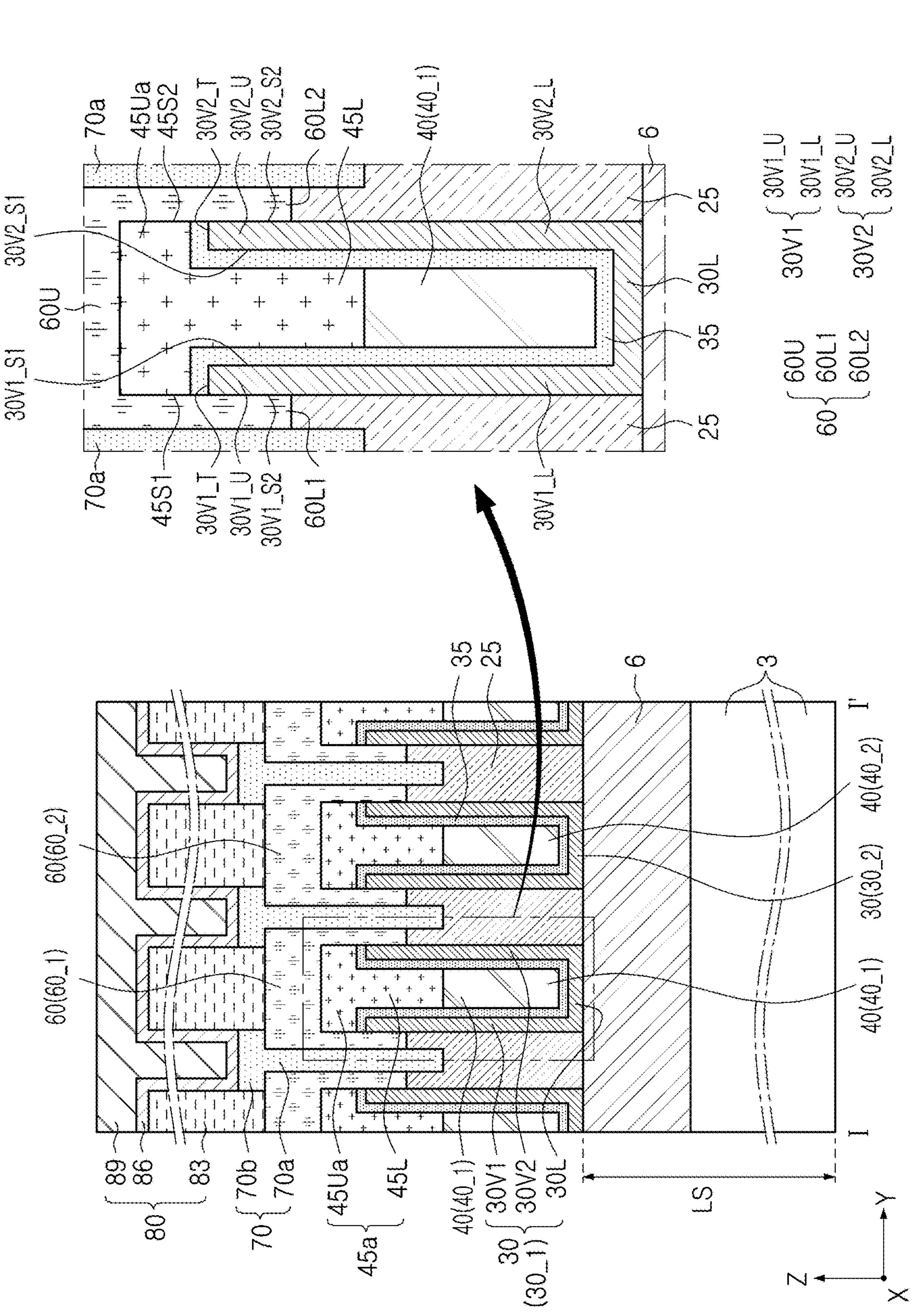
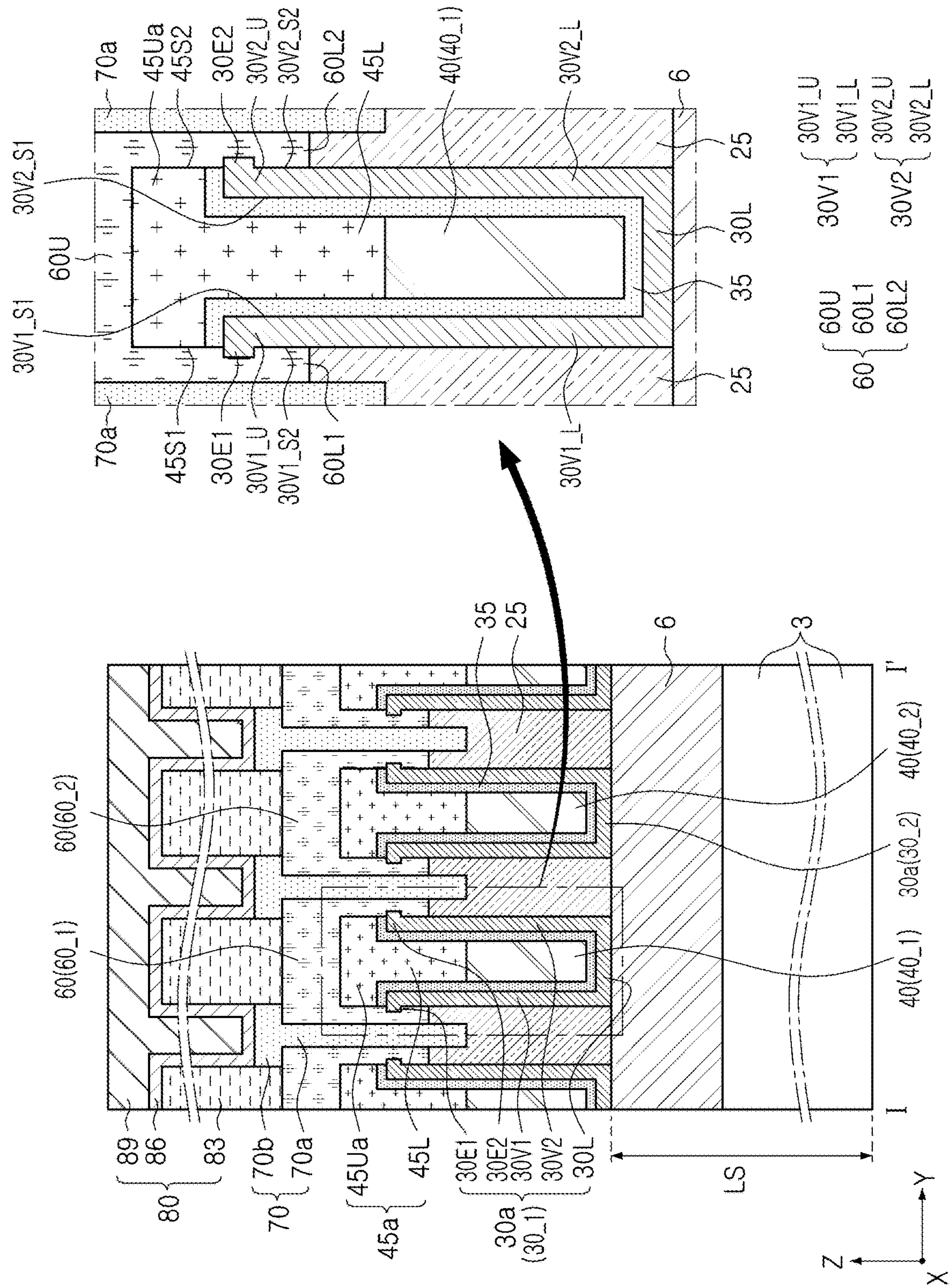

70a
45Ua
45S2
30V2_T
30V2_U
30V2_S2
60L2
45L
40(40_1)
30V2_L
6
30V2_S1
60U
30V1_S1
25
30L
35
70a
45S1
30V1_T
30V1_U
30V1_S2
60L1
30V1_L
30V1_U
30V1_L
30V1
30V2_U
30V2_L
30V2
60U
60L1
60L2
60
60(60_1)
60(60_2)
35
25
6
3
I
I'
40(40_1)
30(30_2)
40(40_2)
89
86
83
80
70b
70a
70
45Ua
45L
45a
40(40_1)
30V1
30V2
30L
30
(30_1)
LS
Z
Y
X

FIG. 4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45Ua
45S2
30E2
30V2_U
30V2_S2
60L2
45L
40(40_1)
30V2_L
6
30V2_S1
60U
30V1_S1
25
30L
35
70a
45S1
30E1
30V1_U
30V1_S2
60L1
30V1_L
30V1_U
30V1_L
30V2_U
30V2_L
30V1
30V2
60U
60L1
60L2
60
89
86
83
70b
70a
45Ua
45L
30E1
30E2
30V1
30V2
30L
80
70
45a
30a
(30_1)
LS
60(60_1)
60(60_2)
40(40_1)
30a(30_2)
40(40_2)
3
I
I'
Z
Y
X

FIG. 5

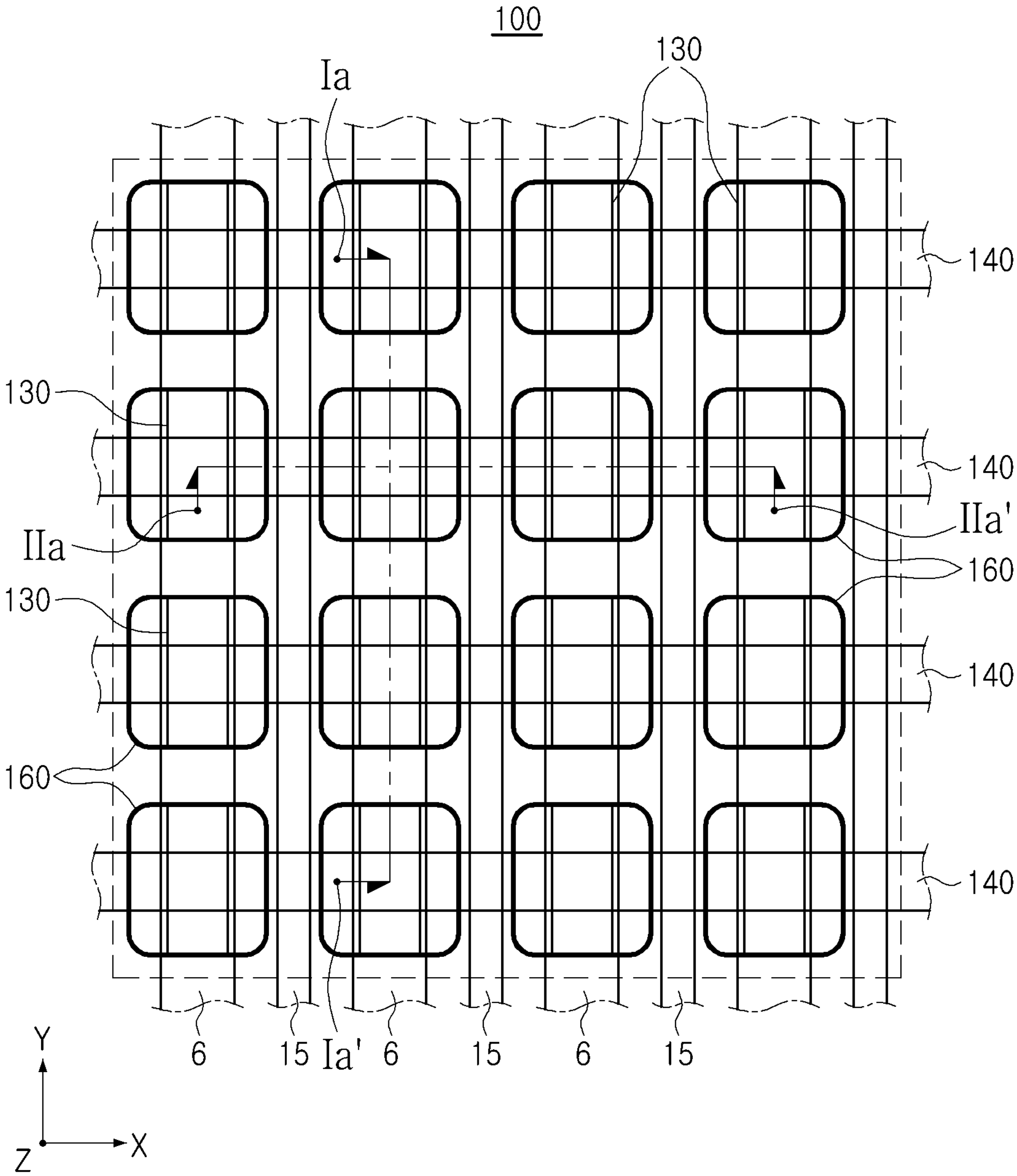



FIG. 6

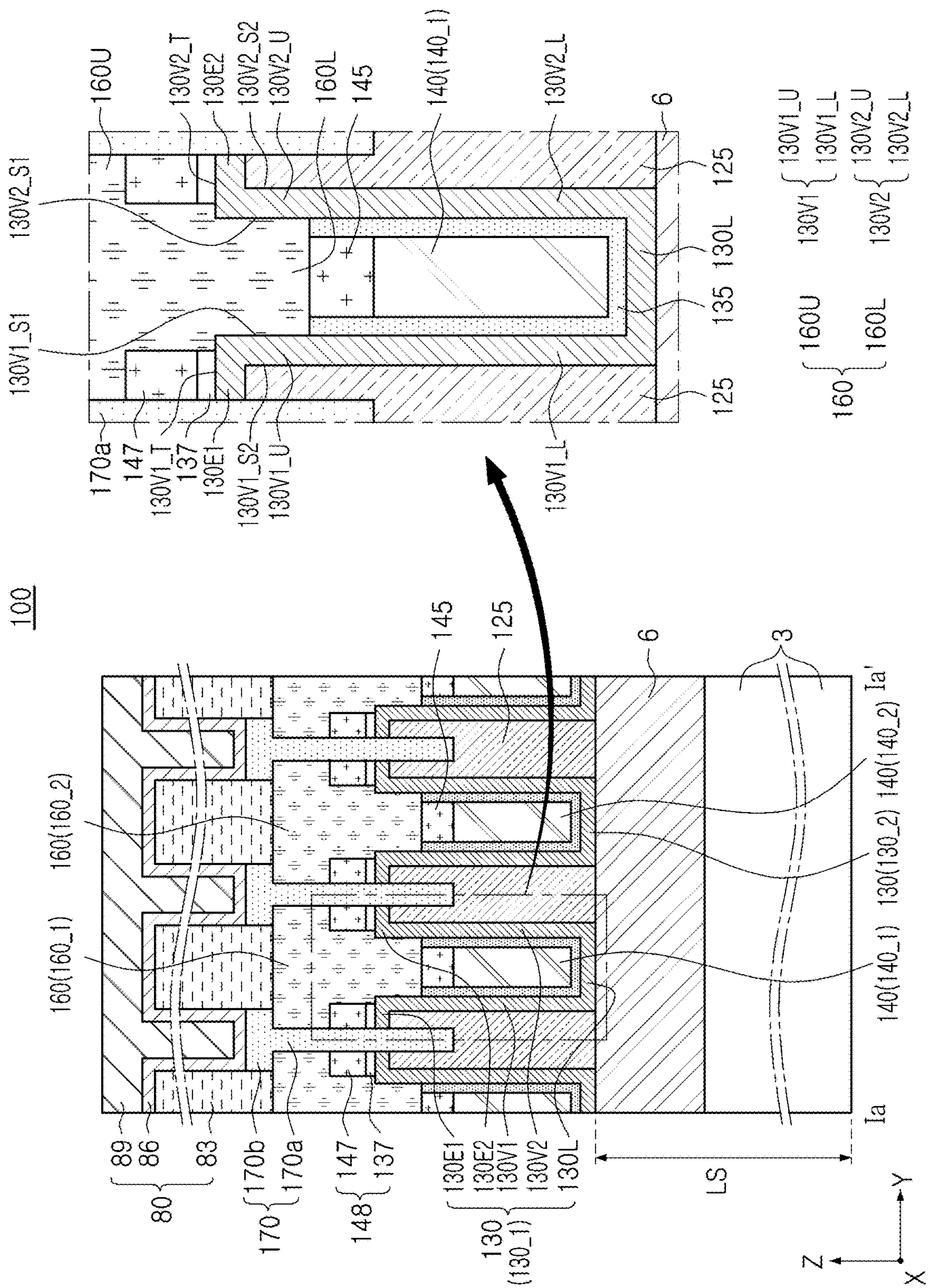
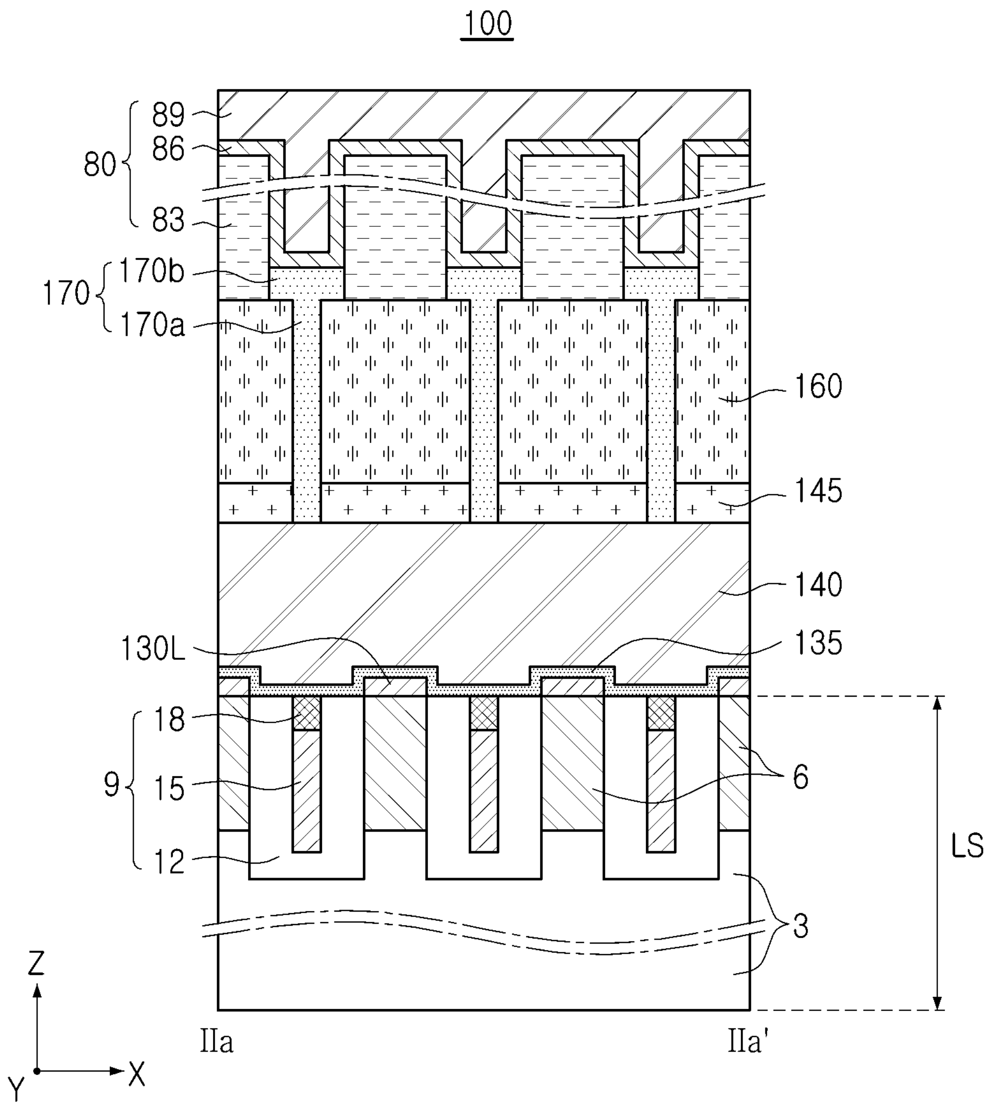

100
80
89
86
83
170
170b
170a
148
147
137
130
(130_1)
130E1
130E2
130V1
130V2
130L
LS
Ia
Ia'
3
6
125
145
160(160_1)
160(160_2)
140(140_1)
140(140_2)
130(130_2)
Z
Y
X
160U
130V2_T
130E2
130V2_S2
130V2_U
160L
145
140(140_1)
130V2_L
130V2_S1
130V1_S1
135
130L
170a
147
130V1_T
137
130E1
130V1_S2
130V1_U
130V1_L
160
130V1
130V1_U
130V1_L
130V2
130V2_U
130V2_L

FIG. 7A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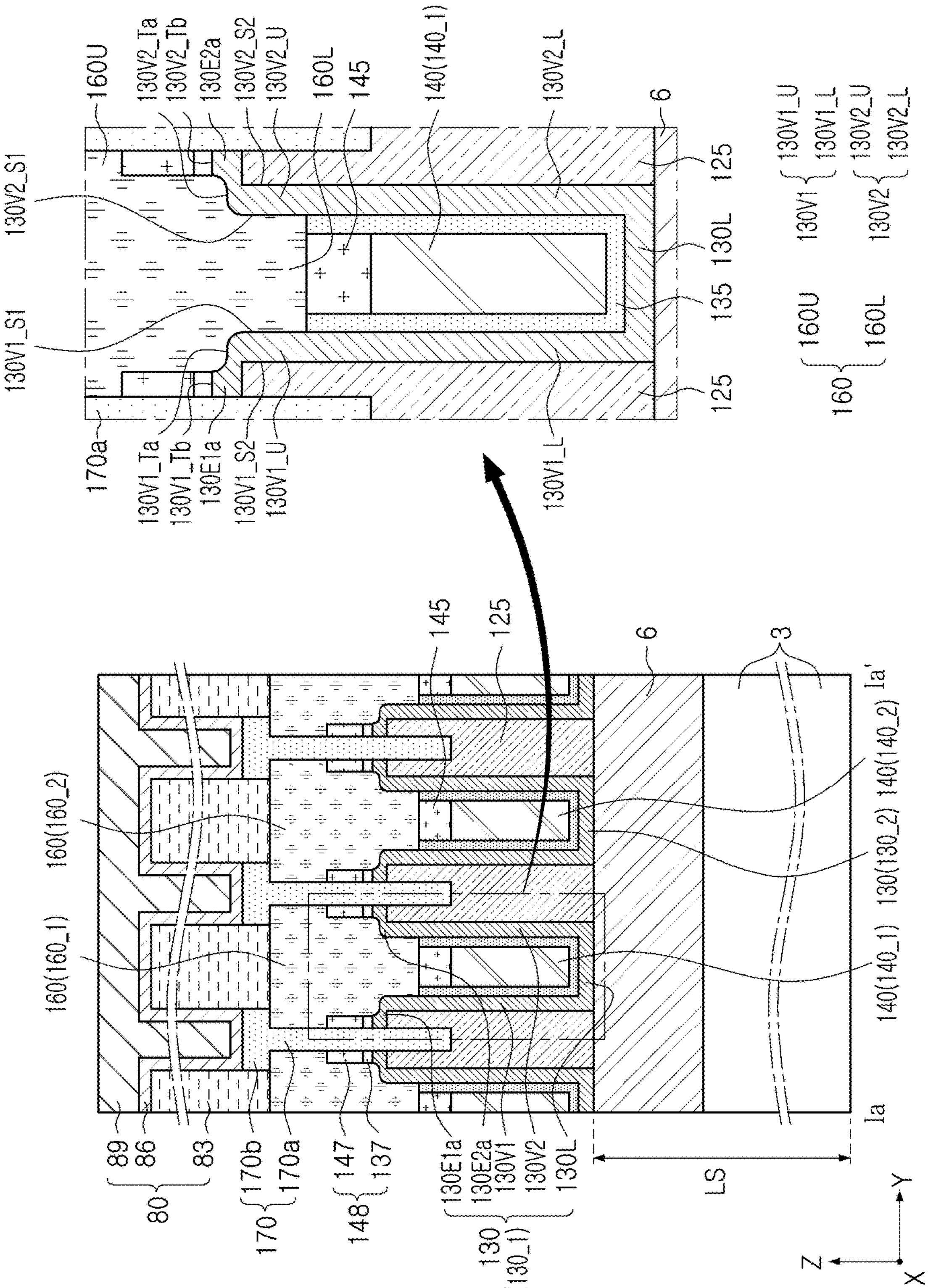
89
86
80
83
170b
170
170a
160
145
140
130L
135
18
9
15
6
12
LS
3
IIa
IIa'
Z
Y
X

FIG. 7B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130V2_Tb
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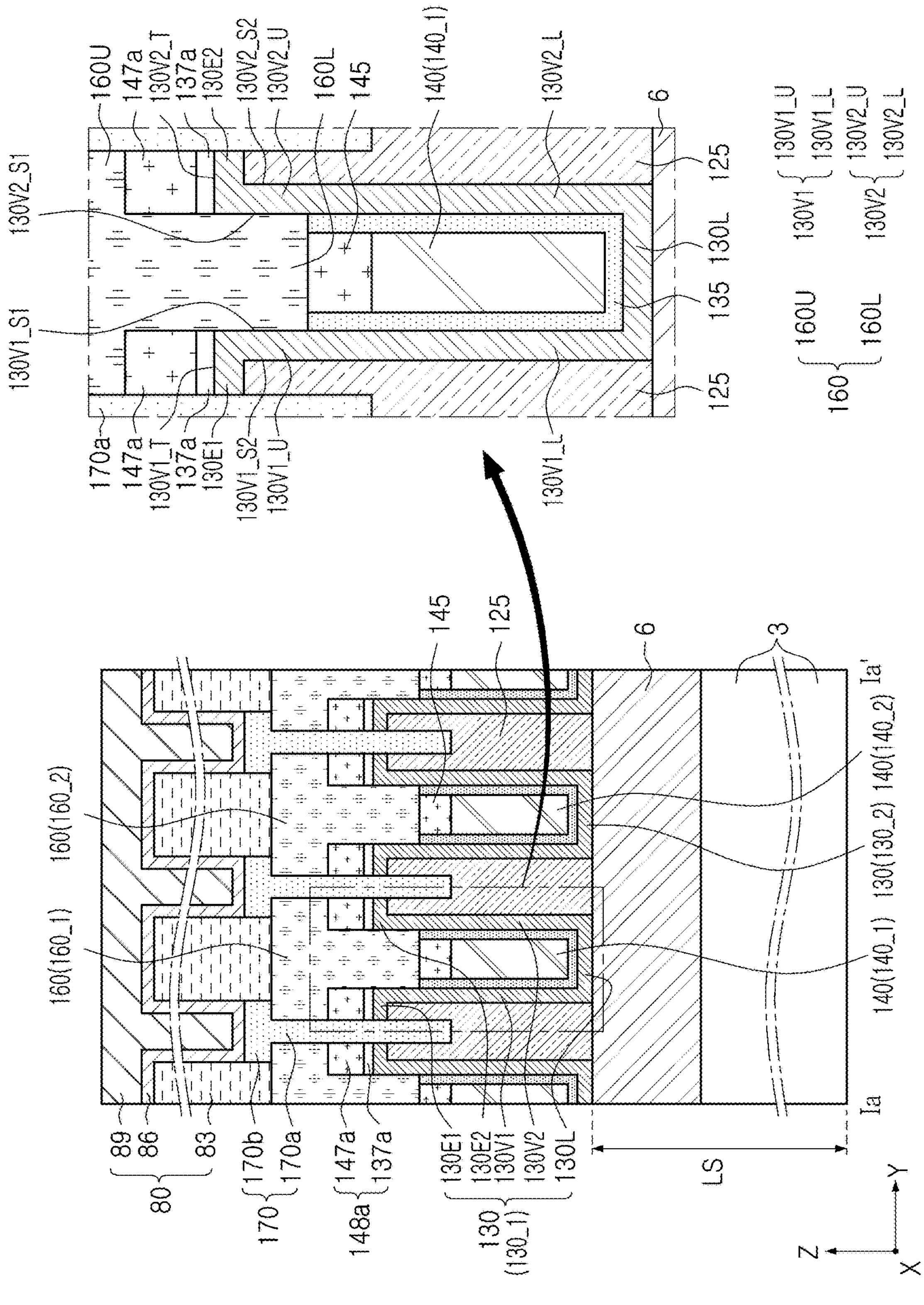
130V2_L
6
125
130L
135
125
170a
130V1_Ta
130V1_Tb
130E1a
130V1_S2
130V1_U
130V1_L
160
160U
160L
130V1
130V1_U
130V1_L
130V2
130V2_U
130V2_L
160(160_1)
160(160_2)
145
125
6
3
80
89
86
83
170
170b
170a
148
147
137
130
(130_1)
130E1a
130E2a
130V1
130V2
130L
LS
Ia
Ia'
140(140_1)
130(130_2)
140(140_2)
Z
Y
X

FIG. 8


160U
147a
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137a
130E2
130V2_S2
130V2_U
160L
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130V2_L
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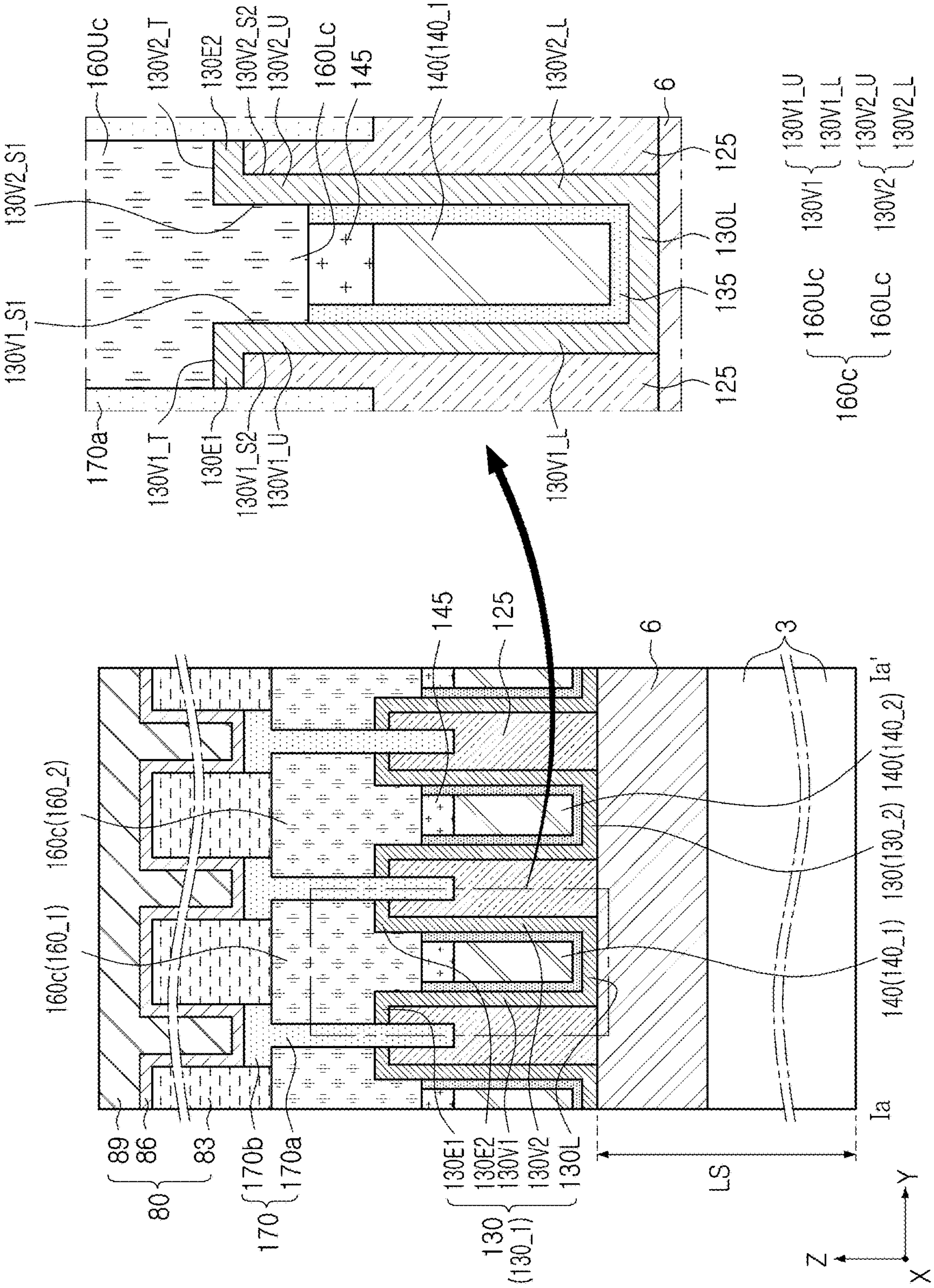
130V1_S1
170a
147a
130V1_T
137a
130E1
130V1_S2
130V1_U
130V1_L
125
135
130L
160
160U
160L
130V1
130V1_U
130V1_L
130V2
130V2_U
130V2_L
160(160_1)
160(160_2)
145
125
6
3
140(140_1)
130(130_2)
140(140_2)
Ia
Ia'
80
89
86
83
170
170b
170a
148a
147a
137a
130
(130_1)
130E1
130E2
130V1
130V2
130L
LS
Z
Y
X

FIG. 9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130E2
130V2_S2
130V2_U
160Lc
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130V2_L
6
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130V1_S1
170a
130V1_T
130E1
130V1_S2
130V1_U
130V1_L
125
135
130L
160c
160Uc
160Lc
130V1
130V1_U
130V1_L
130V2
130V2_U
130V2_L
160c(160_1)
160c(160_2)
145
125
6
3
Ia'
Ia
140(140_1)
130(130_2)
140(140_2)
80
89
86
83
170
170b
170a
130
(130_1)
130E1
130E2
130V1
130V2
130L
LS
Z
Y
X

FIG. 10

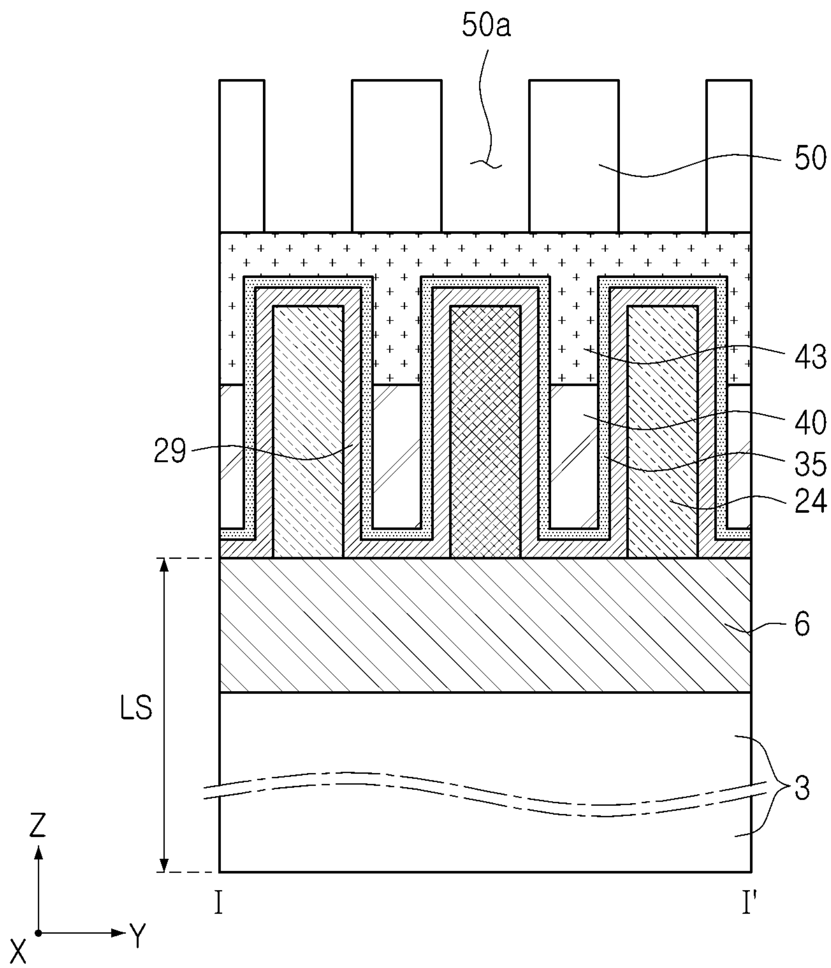


FIG. 11A

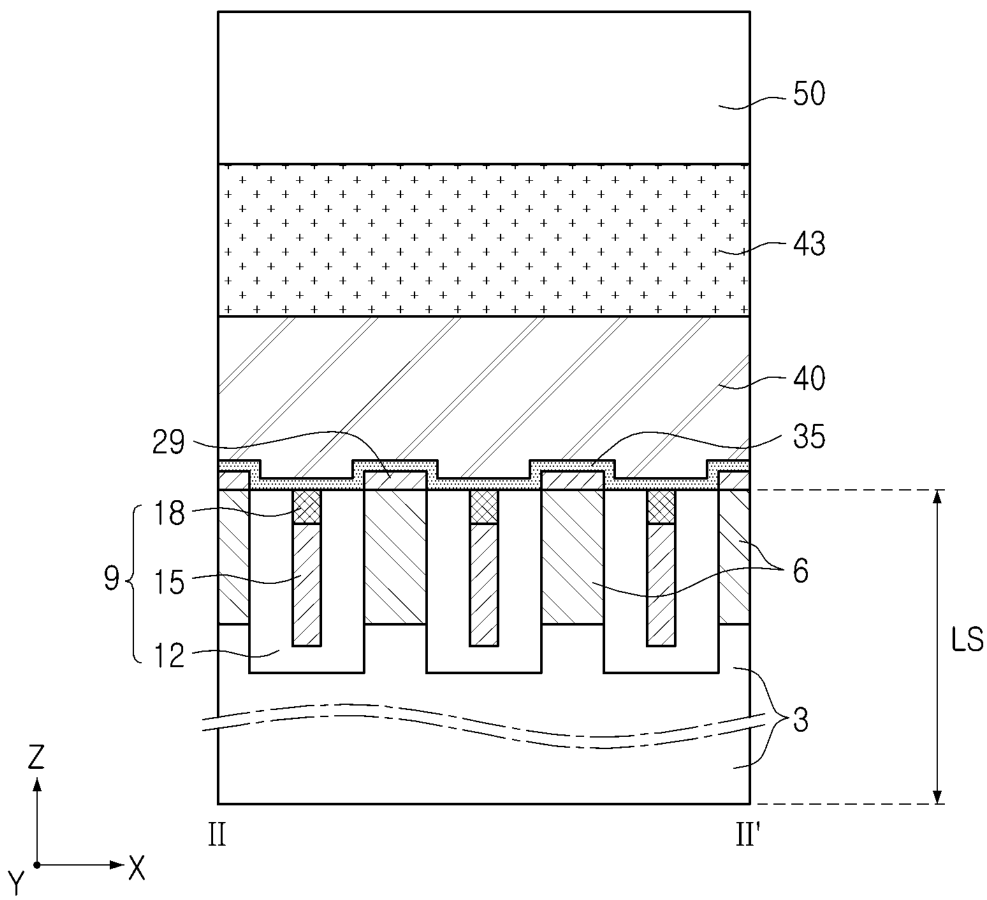


FIG. 11B

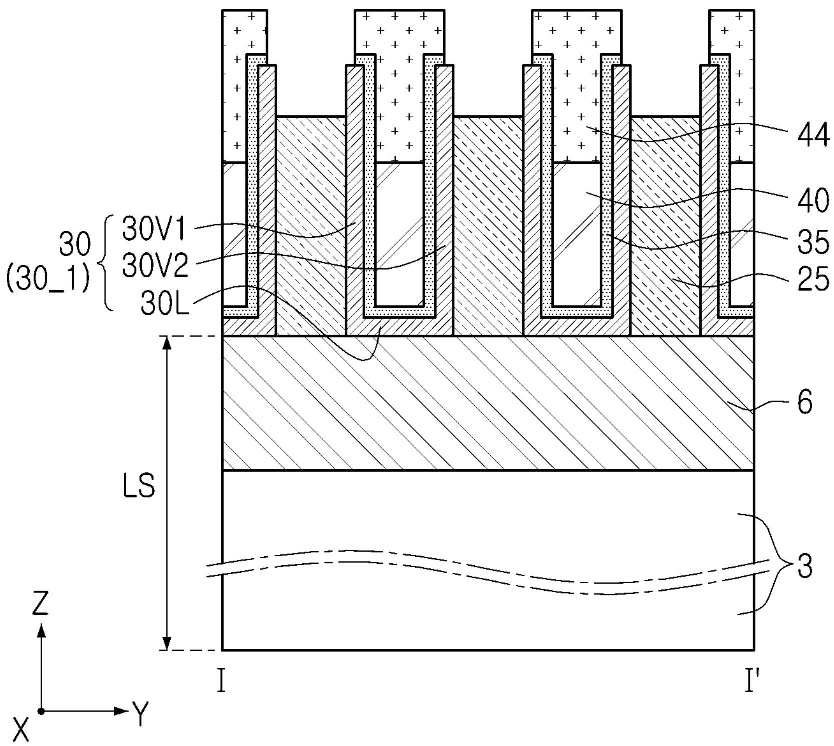


FIG. 12

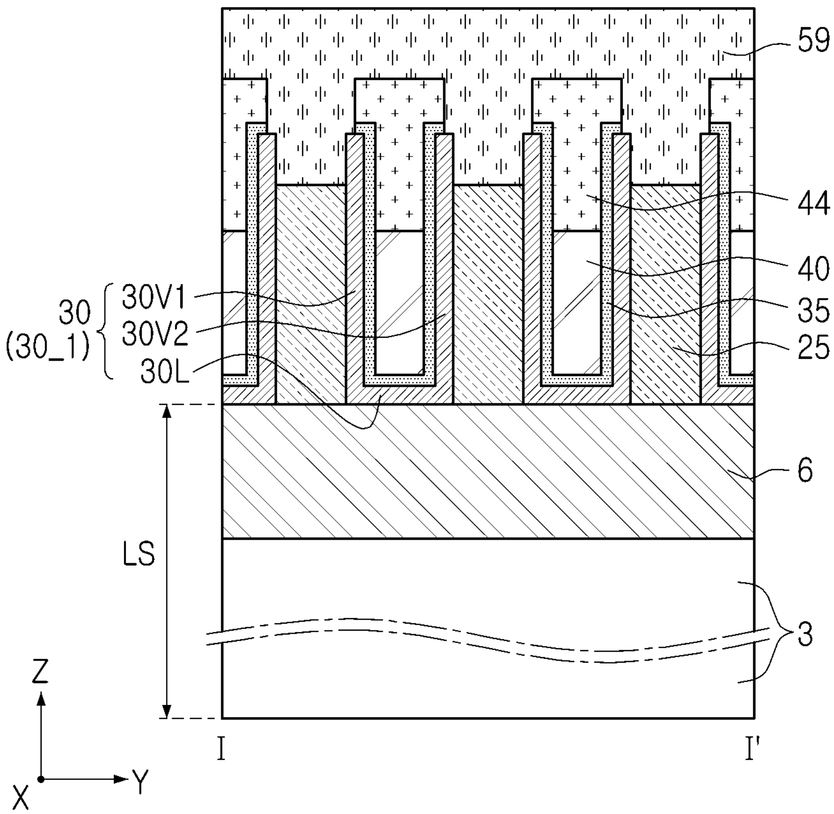


FIG. 13A

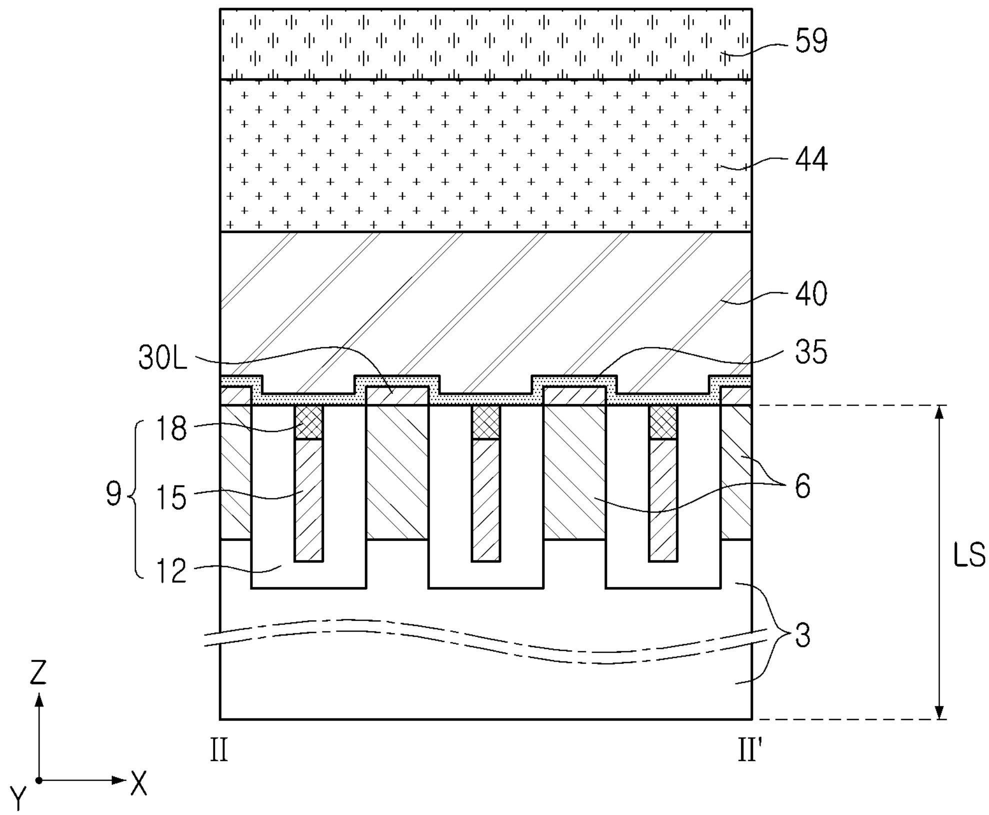


FIG. 13B

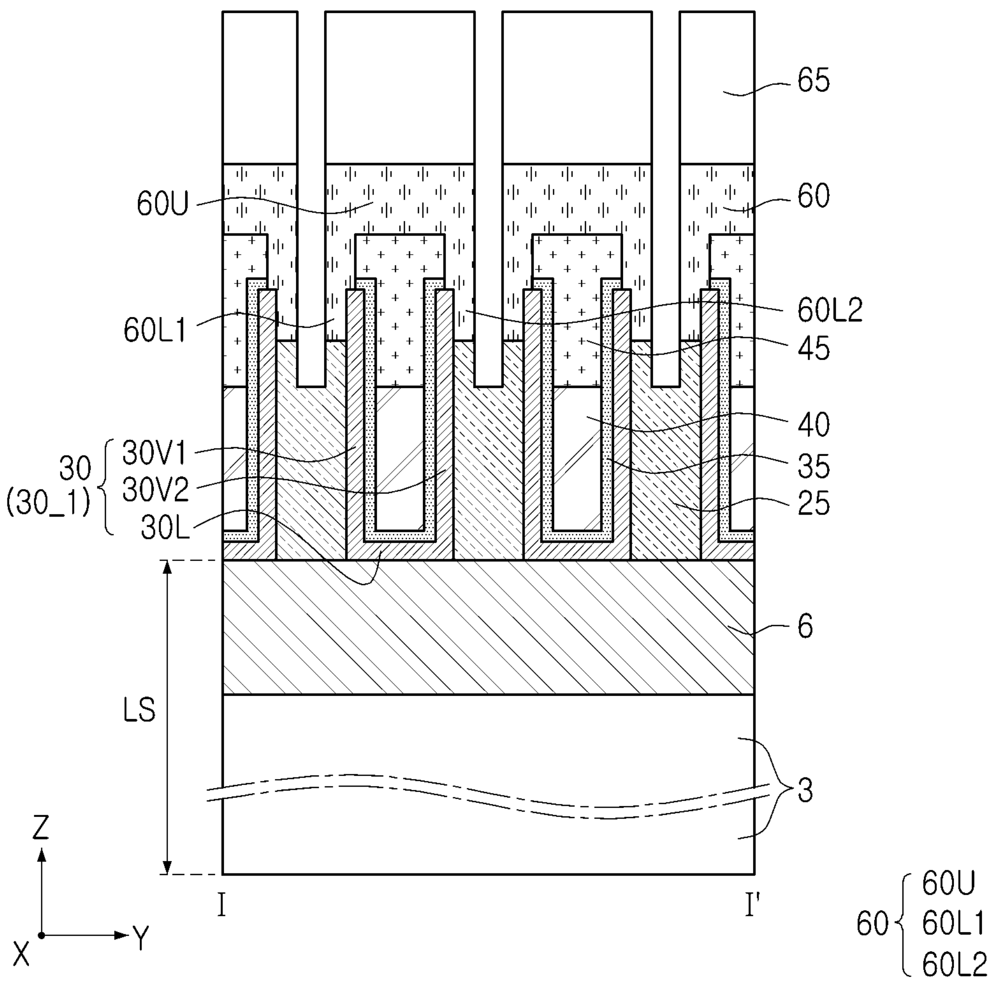
65
60U
60
60L1
60L2
45
40
30
(30_1)
30V1
30V2
30L
35
25
6
LS
3
Z
Y
X
I
I'
60
60U
60L1
60L2

FIG. 14A

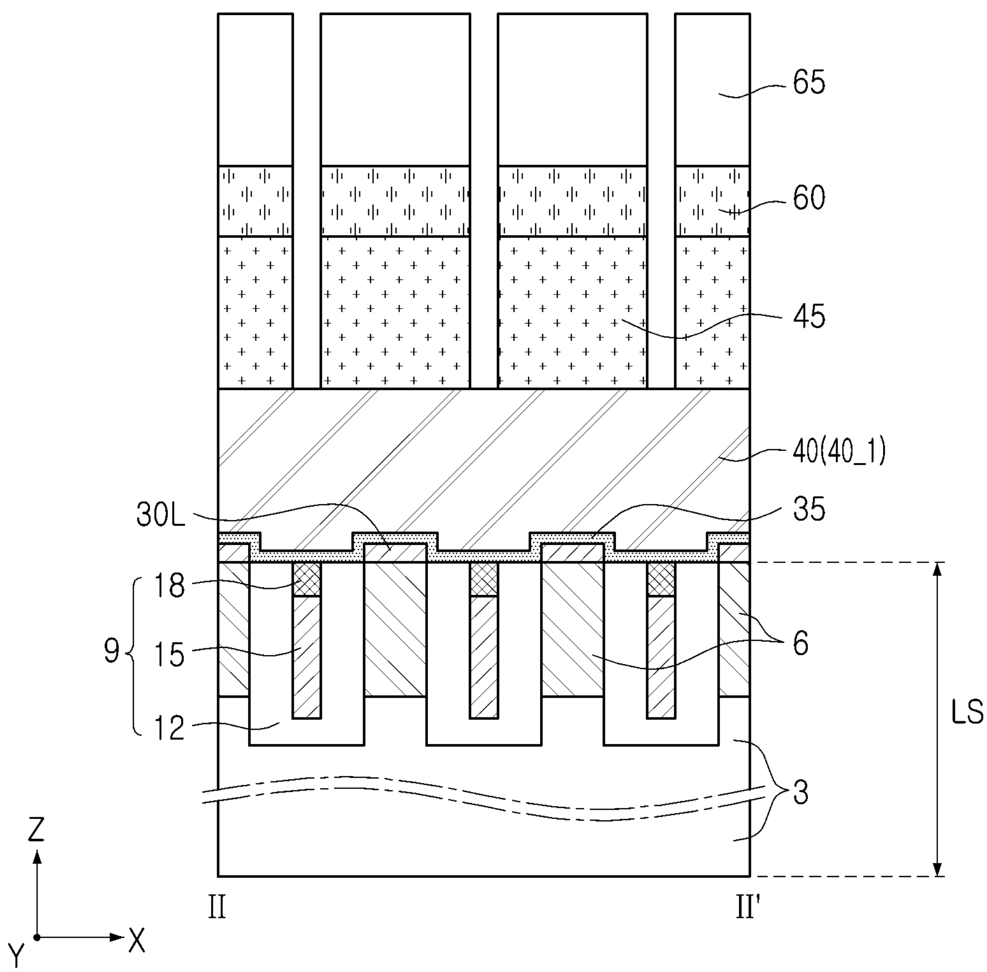


FIG. 14B

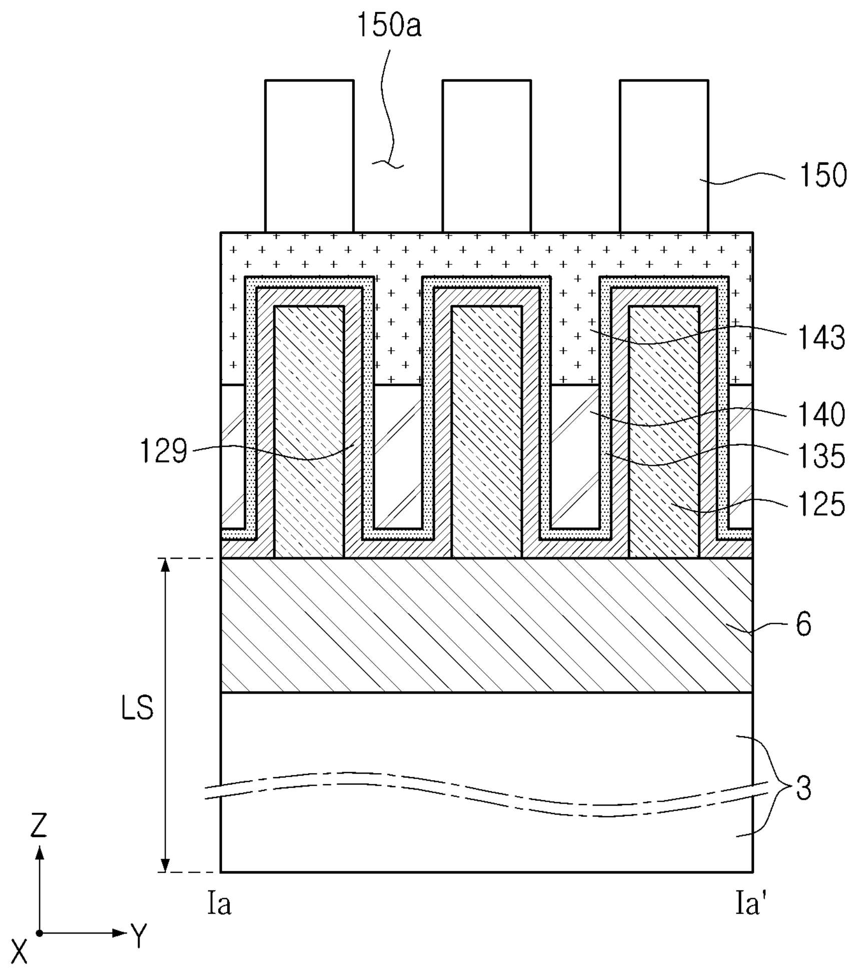


FIG. 15A

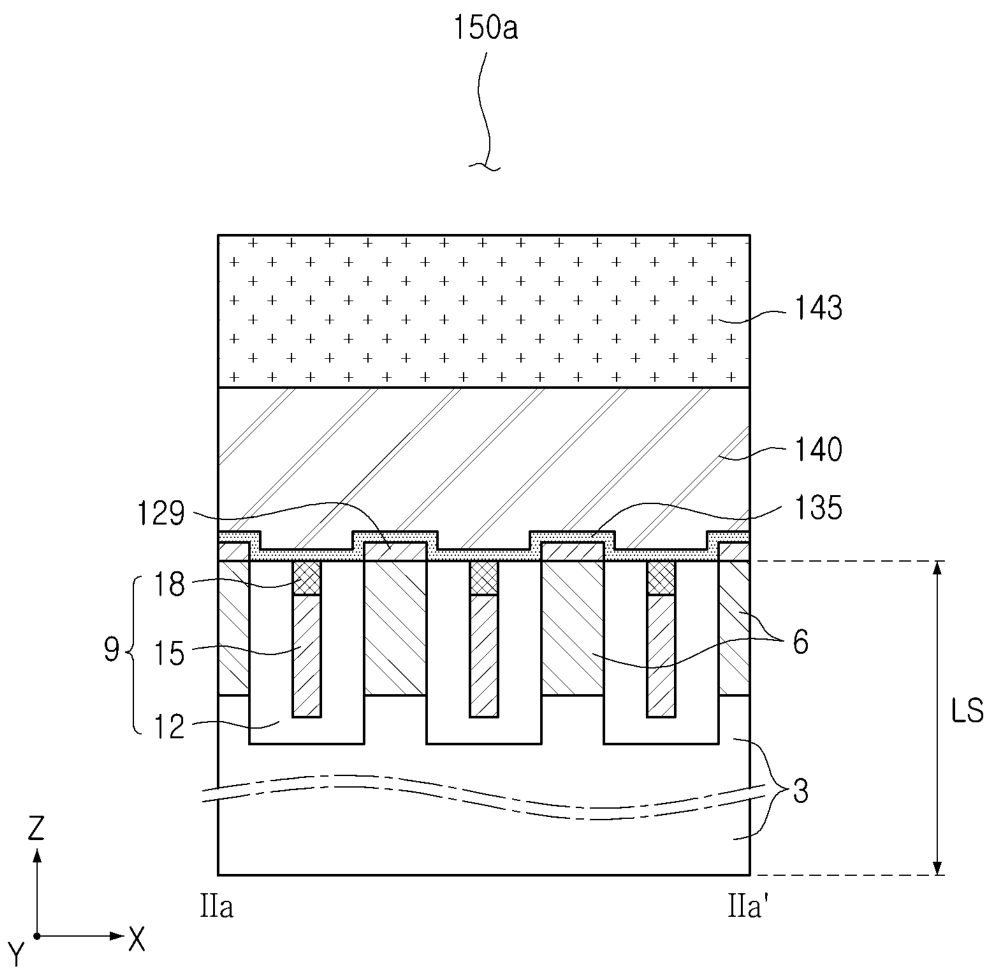


FIG. 15B

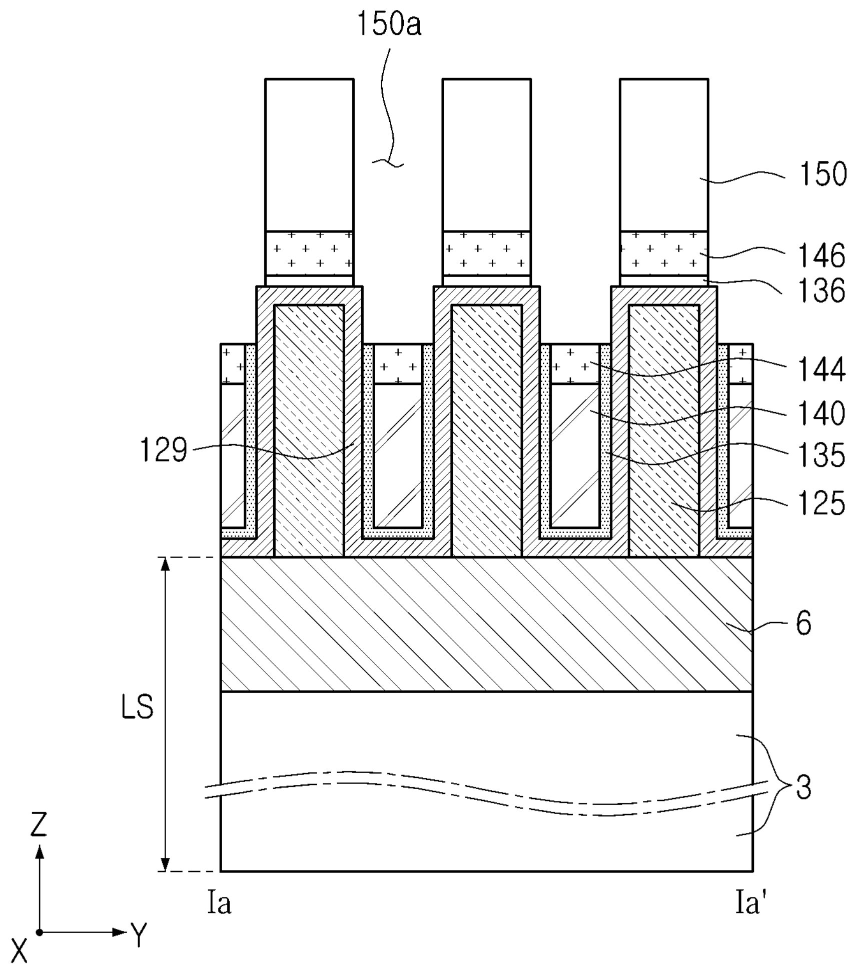


FIG. 16A

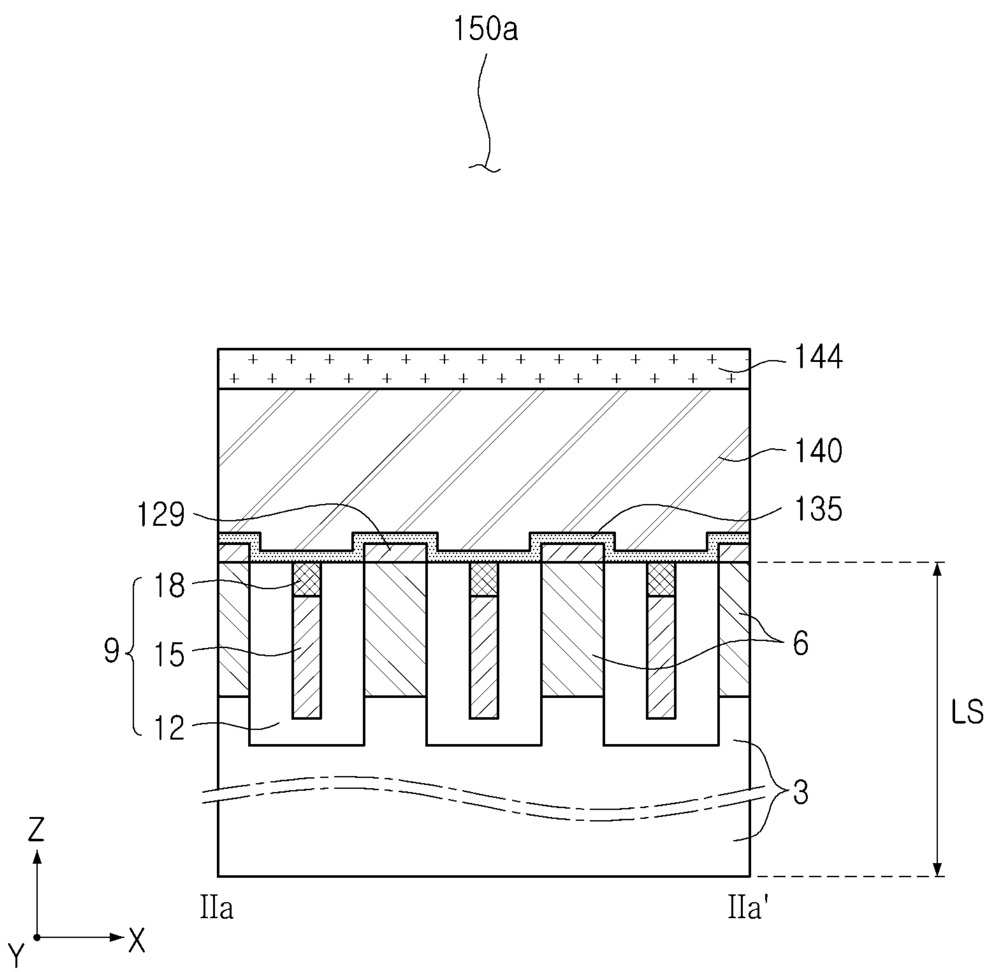


FIG. 16B

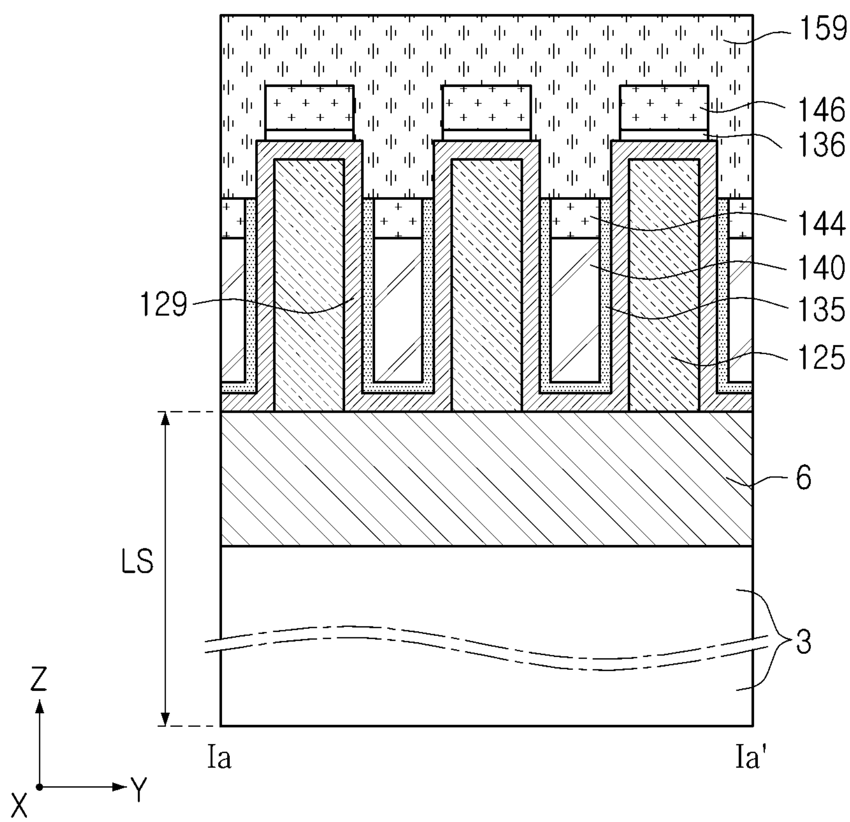


FIG. 17A

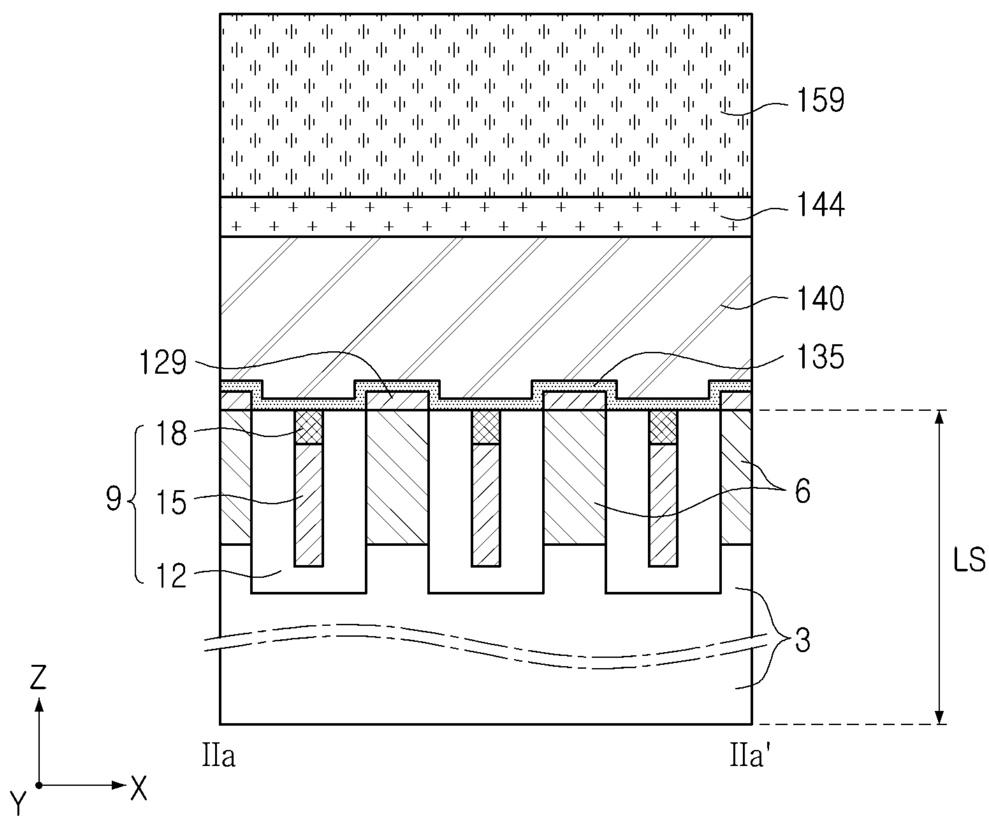


FIG. 17B

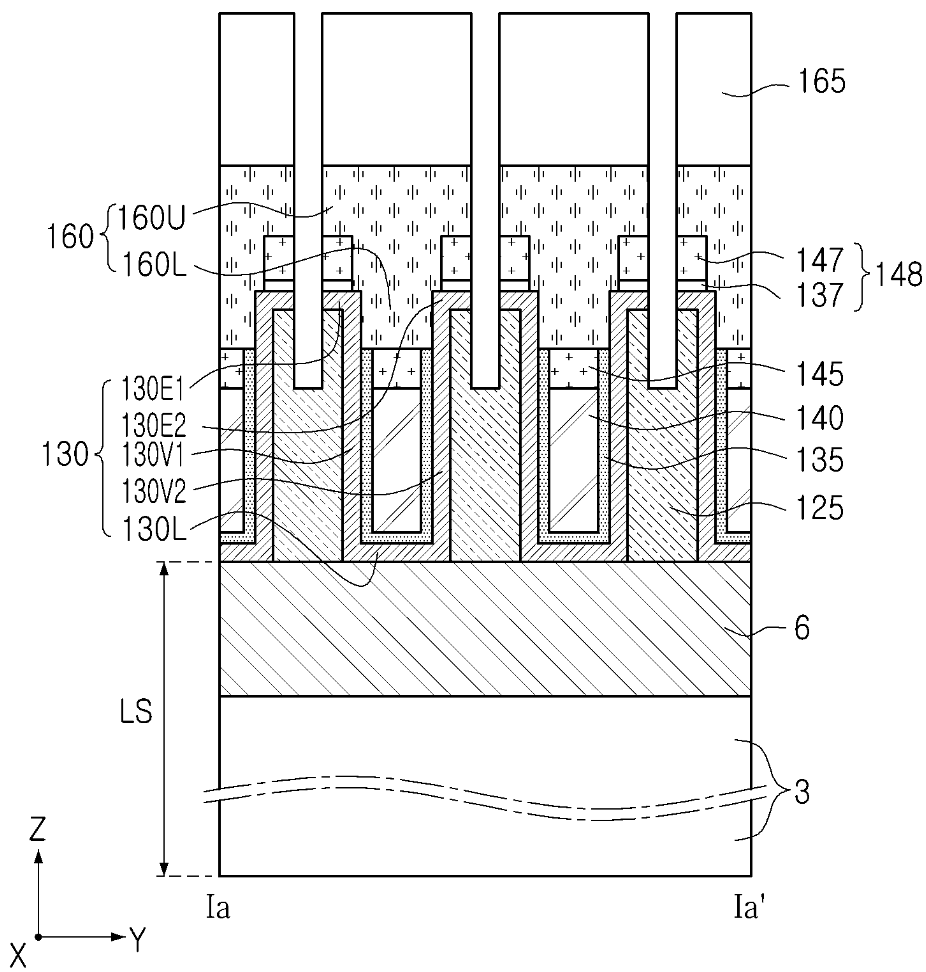
165
160
160U
160L
147
137
148
130E1
130E2
130V1
130V2
130L
130
145
140
135
125
6
LS
3
Z
X
Y
Ia
Ia'

FIG. 18A

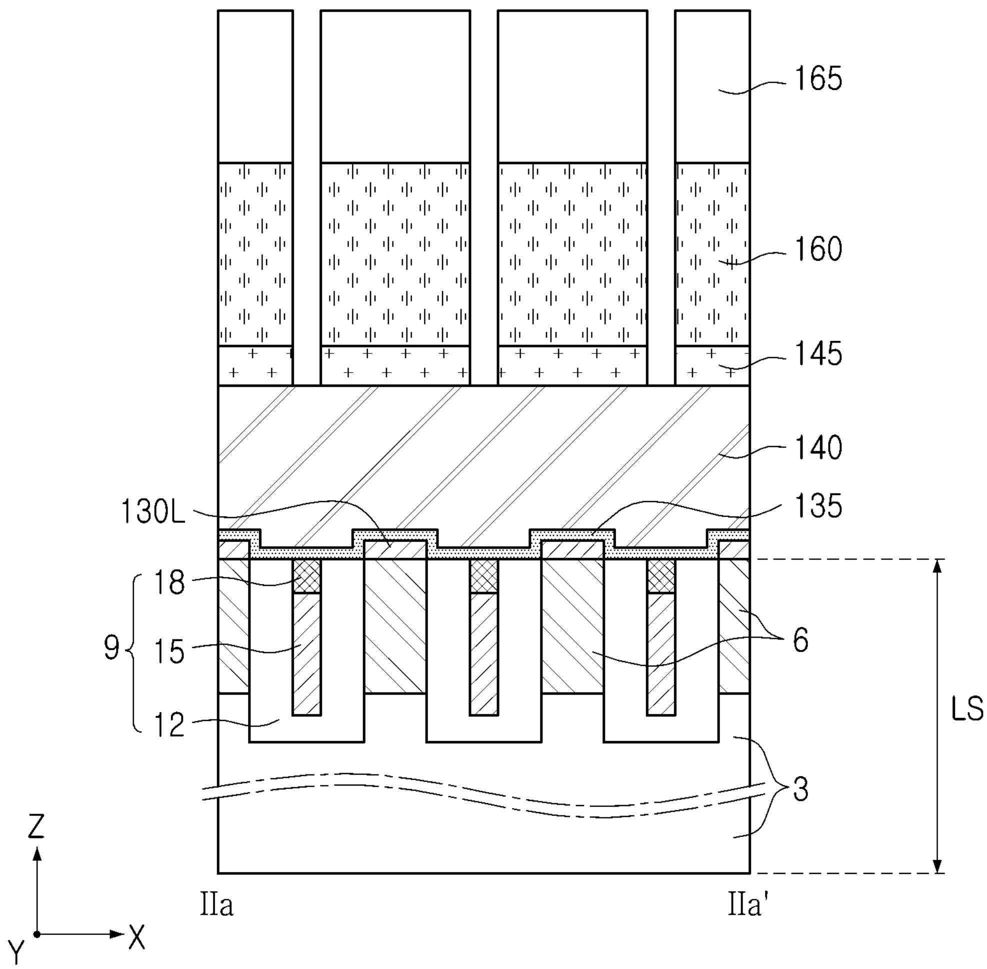


FIG. 18B

# SEMICONDUCTOR DEVICE INCLUDING ACTIVE PATTERN AND CONDUCTIVE PAD PATTERN

## CROSS-REFERENCE TO RELATED APPLICATION(S)

**[0001]** This application claims the benefit under 35 USC 119(a) of Korean Patent Application No. 10-2024-0032012 filed on Mar. 6, 2024 in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

## BACKGROUND

**[0002]** The present inventive concept relates to a semiconductor device including an active pattern and a conductive pad pattern electrically connected and a method of forming the same.

**[0003]** Research is underway to reduce the size and improve the performance of elements in semiconductor devices. For example, in DRAMs, research is underway to reliably and stably form elements having a reduced size, but as the size of the elements is reduced, dispersion characteristics of semiconductor devices are deteriorating.

## SUMMARY

**[0004]** Example embodiments provide a semiconductor device in which integration may be increased and performance may be improved.

**[0005]** Example embodiments provide a method of forming the semiconductor device.

**[0006]** According to example embodiments, a semiconductor device is provided. The semiconductor device includes a bit line, an active pattern on the bit line, and including a lower active portion electrically connected to the bit line, a first vertical active portion extending from the lower active portion, and a second vertical active portion extending from the lower active portion, with the first vertical active portion and the second vertical active portion are spaced apart from one another, a word line between a lower region of the first vertical active portion and a lower region of the second vertical active portion, and a conductive pad pattern electrically connected to an upper region of the first vertical active portion and an upper region of the second vertical active portion.

**[0007]** According to example embodiments, a semiconductor device is provided. The semiconductor device includes a conductive line extending in a first horizontal direction, a first active pattern and a second active pattern on the conductive line, a first gate electrode and a second gate electrode on a level higher than a level of the conductive, and a first conductive pad pattern and a second conductive pad pattern on the first and second gate electrodes, respectively. Each of the first and second active patterns includes a lower active portion electrically connected to an upper surface of the conductive line, a first vertical active portion extending from the lower active portion, and a second vertical active portion extending from the lower active portion. The first gate electrode is between the first vertical active portion of the first active pattern and the second vertical active portion of the first active pattern, and is on the lower active portion of the first active pattern. The second gate electrode is between the first vertical active portion of the second active pattern and the second vertical active portion of the second active pattern, and is on the lower active portion of the first active pattern. The first conductive pad pattern is electrically connected to the first vertical active portion of the first active pattern and the second vertical active portion of the first active pattern. The second conductive pad pattern is electrically connected to the first vertical active portion of the second active pattern and the second vertical active portion of the second active pattern.

**[0008]** According to example embodiments, a semiconductor device is provided. A semiconductor device includes a conductive line extending in a first horizontal direction, a cell transistor on the conductive line, and a conductive pad pattern on the cell transistor. The cell transistor includes an active pattern including a first vertical channel portion and a second vertical channel portion spaced apart from each other in the first horizontal direction, a single gate electrode between the first and second vertical channel portions, and a gate dielectric layer between the single gate electrode and the active pattern. The conductive pad pattern contacts upper regions of the first and second vertical channel portions.

## BRIEF DESCRIPTION OF DRAWINGS

**[0009]** The above and other aspects, features, and advantages of the present inventive concept will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

**[0010]** FIGS. **1**, **2**A, and **2**B are diagrams illustrating a semiconductor device according to example embodiments;

**[0011]** FIG. **3** is a cross-sectional view illustrating a modified example of a semiconductor device according to example embodiments;

**[0012]** FIG. **4** is a cross-sectional view illustrating a modified example of a semiconductor device according to example embodiments;

**[0013]** FIG. **5** is a diagram illustrating a modified example of a semiconductor device according to example embodiments;

**[0014]** FIGS. **6**, **7**A, and **7**B are diagrams illustrating modified examples of a semiconductor device according to example embodiments;

**[0015]** FIG. **8** is a cross-sectional view illustrating a modified example of a semiconductor device according to example embodiments;

**[0016]** FIG. **9** is a cross-sectional view illustrating a modified example of a semiconductor device according to example embodiments;

**[0017]** FIG. **10** is a cross-sectional view illustrating a modified example of a semiconductor device according to example embodiments;

**[0018]** FIGS. **11**A, **11**B, **12**, **13**A, **13**B, **14**A, and **14**B are cross-sectional views illustrating an example of a method of forming a semiconductor device according to example embodiments; and

**[0019]** FIGS. **15**A, **15**B, **16**A, **16**B, **17**A, **17**B, **18**A, and **18**B are cross-sectional views illustrating another example of a method of forming a semiconductor device according to example embodiments.

## DETAILED DESCRIPTION

**[0020]** Hereinafter, terms such as "upper," "intermediate," "middle," "lower" and the like may be replaced with other terms, such as "first," "second," "third" and the like to describe the elements of the specification. Terms such as "first," "second" and "third" may be used to describe various elements, but the elements are not limited by the terms, and a "first element" may be referred to as a "second element." In the specification, terms such as 'lower', 'upper', 'upper end', and 'lower end' may be terms that are described based on the drawings.

**[0021]** In the drawings, reference numerals displayed in formats such as "**30**(**30_1**)", "**30**(**30_2**)", and the like may mean that the element of reference numeral **30** may be plural, and a plurality of elements represented by reference numeral **30** may include a first element represented by symbol **30_1** and a second element represented by symbol **30_2**.

**[0022]** First, an illustrative example of a semiconductor device according to example embodiments will be described with reference to FIGS. **1**, **2**A, and **2**B. In FIGS. **1** to **2**B, FIG. **1** is a plan view illustrating a semiconductor device according to example embodiments, FIG. **2**A is a cross-sectional view conceptually illustrating the area taken along line I-I' of FIG. **1**, and FIG. **2**B is a partial enlarged view of the area indicated by II-II' in FIG. **1**.

**[0023]** Referring to FIGS. **1**, **2**A, and **2**B, the semiconductor device **1** according to example embodiments may include a lower structure LS.

**[0024]** The lower structure LS may include a base **3** and conductive lines **6** on the base. The base **3** may include core circuits and peripheral circuits in memory such as DRAM. Each of the conductive lines **6** may have a line shape extending in the first horizontal direction Y. The conductive lines **6** may be spaced apart from each other in a second horizontal direction X perpendicular to the first horizontal direction Y. Each of the conductive lines **6** may include at least one conductive material layer. For example, each of the conductive lines **6** may include doped polysilicon, metal, conductive metal nitride, metal-semiconductor compound, conductive metal oxide, conductive graphene, carbon nanotube, or combinations thereof. For example, each of the conductive lines **6** may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, graphene, carbon nanotube, or combinations thereof, but is not limited thereto. Each of the conductive lines **6** may comprise a single layer or multiple layers of the materials described above.

**[0025]** The conductive lines **6** may be bit lines **6** in a memory such as DRAM. Hereinafter, the conductive lines **6** will be described and referred to as 'bit lines.'

**[0026]** The lower structure LS may further include shield structures **9** disposed between the bit lines **6**. One of the shield structures **9** may be disposed between a pair of adjacent bit lines among the bit lines **6**. Each of the shield structures **9** may include a shield conductive line **15**, a shield insulating capping pattern **18** on the shield conductive line **15**, and a shield insulating layer **12** covering, overlapping, or on the side surfaces of the shield conductive line **15** and the shield insulating capping pattern **18** and covering, overlapping, or on the lower surface of the shield conductive line **15**, stacked sequentially. The shield conductive lines **15** may have a lower surface disposed at a lower level than level of the lower surface of the bit lines **6**, and an upper surface disposed at a higher level than level of the lower surface of the bit lines **6** and a lower level than level of the upper surface of the bit lines **6**. The shield conductive line **15** may screen capacitive coupling between the bit lines **6**. For example, the shield conductive line **15** may significantly reduce the Resistive-Capacitive delay (RC delay) of the bit lines **6** by reducing or blocking parasitic capacitance between the bit lines **6**. The shield conductive line **15** is formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaiN, RuTiN, NiSi, CoSi, IrOx, RuOx, graphene, carbon nanotubes, or combinations thereof, but is not limited thereto. The shield conductive line **15** may include a single layer or multiple layers of the materials described above.

**[0027]** The semiconductor device **1** may further include insulating patterns **25** disposed on the lower structure LS. Each of the insulating patterns **25** may have a line shape extending in the second horizontal direction X. The insulating patterns **25** may be formed of an insulating material such as silicon oxide.

**[0028]** The semiconductor device **1** may further include active patterns **30** arranged in the first horizontal direction Y and the second horizontal direction X. The active patterns **30** may be semiconductor patterns or channel patterns. The active patterns **30** may include a semiconductor material. For example, the active patterns **30** may include an oxide semiconductor. The oxide semiconductor may be indium gallium zinc oxide (IGZO). However, the example embodiments are not limited thereto. For example, the oxide semiconductor layer may include at least one of Indium Tungsten Oxide (IWO), Indium Tin Gallium Oxide (ITGO), Indium Aluminum Zinc Oxide (IAGO), Indium Gallium Oxide (IGO), Indium Tin Zinc Oxide (ITZO), zinc tin oxide (ZTO), indium zinc oxide (IZO), ZnO, Iindium gallium silicon oxide (GSO), indium oxide (InO), tin oxide (SnO), titanium oxide (TiO), zinc oxynitride (ZnON), magnesium zinc oxide (MgZnO), indium zinc oxide (InZnO)), indium gallium zinc oxide (InGaZnO), zirconium indium zinc oxide (ZrInZnO), hafnium indium zinc oxide (HfInZnO), tin indium zinc oxide (SnInZnO), aluminum tin indium zinc oxide (AlSnInZnO), silicon indium zinc oxide (SiInZnO), zinc tin oxide (ZnSnO), aluminum zinc tin oxide (AlZnSnO), gallium zinc tin oxide (GaZnSnO), zirconium zinc tin oxide (ZrZnSnO), and/or Indium Gallium Silicon Oxide (InGaSiO). The active patterns **30** may include the oxide semiconductor, but the example embodiments are not limited thereto. For example, the active patterns **30** may include a semiconductor material such as single crystal silicon or polysilicon.

**[0029]** The active patterns **30** may include a first active pattern **30_1** and a second active pattern **30_2** adjacent to each other in the first horizontal direction Y. Each of the active patterns **30** may include a lower active portion **30**L disposed on the bit line **6** and connected to the bit line **6**, a first vertical active portion **30V1** extending upwardly from the first side of the lower active portion **30**L, and a second vertical active portion **30V2** extending upwardly from the second side of the lower active portion **30**L.

**[0030]** In each of the active patterns **30**, the lower active portion **30**L may contact the upper surface of the bit line **6**. In each of the active patterns **30**, the lower surface of the lower active portion **30**L may contact the upper surface of the bit line **6**. In each of the active patterns **30**, the first vertical active portion **30V1** and the second vertical active portion **30V2** may be arranged in the first horizontal direction Y. In each of the active patterns **30**, the first vertical active portion **30V1** and the second vertical active portion **30V2** may be spaced apart from each other in the first horizontal direction Y. In each of the active patterns **30**, the first vertical active portion **30V1** may include a first lower region **30V1**_L and a first upper region **30V1**_U on the first lower region **30V1**_L, and the second vertical active portion **30V2** may include a first lower region **30V2**_L and a first upper region **30V2**_U on the first lower region **30V2**_L. Upper surfaces of the first and second vertical active portions **30V1** and **30V2** may be disposed at a higher level than level of the upper surfaces of the insulating patterns **25**. The first and second vertical active portions **30V1** and **30V2** may also be referred to as vertical channel portions or vertical channel layers.

**[0031]** The semiconductor device **1** may further include gate electrodes **40**. The gate electrodes **40** may be word lines in memory such as DRAM. Hereinafter, the gate electrodes **40** will be described and referred to as 'word lines.'

**[0032]** The word lines **40** may be disposed at a lower level than level of the upper surfaces of the first and second vertical active portions **30V1** and **30V2**. The word lines **40** may be placed at a lower level than level of the upper surfaces of the insulating patterns **25**. Each of the word lines **40** may have a line shape extending in the second horizontal direction X. The word lines **40** may be spaced apart from each other in the first horizontal direction Y. The word lines **40** may include a first word line **40_1** and a second word line **40_2** that are adjacent to each other in the first horizontal direction Y. The word lines **40** may be disposed on the lower active portions **30**L and between the first vertical active portions **30V1** and the second vertical active portions **30V2**. One of the word lines **40** may be disposed on one of the active patterns **30**. For example, the first word line **40_1** may be disposed on the lower active portion **30**L of the first active pattern **30_1**, and may be disposed between the first vertical active portion **30V1** of the first active pattern **30_1** and the second vertical active portion **30V2** of the first active pattern **30 1**. The first word line **40_1** may be disposed between the lower region **30V1**_L of the first vertical active portion **30V1** and the lower region **30V2**_L of the second vertical active portion **30V2** of the first active pattern **30_1**. Each of the word lines **40** may include at least one conductive material. For example, each of the word lines **40** may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, or combinations thereof, but is not limited thereto. Each of the word lines **40** may include a single layer or multiple layers of the materials described above.

**[0033]** The semiconductor device **1** may further include insulating capping patterns **45** on the word lines **40**. The insulating capping patterns **45** may be formed of an insulating material such as silicon nitride or silicon oxynitride. Each of the insulating capping patterns **45** may include an upper capping portion **45**U disposed at a higher level than level of the active pattern **30**, and a lower capping portion **45**L extending downwardly (i.e., towards the base **3**) from the center of the upper capping portion **45**U and contacting the upper surface of the word line **40**. The upper capping portion **45**U may include at least one vertically overlapping portion of the first and second vertical active portions **30V1** and **30V2**.

**[0034]** The semiconductor device **1** may further include gate dielectric layers **35** disposed between the active patterns **30** and the word lines **40** and between the active patterns **30** and the insulating capping patterns **45**. The gate dielectric layers **35** may be in contact with the active patterns **30** and the word lines **40**, between the active patterns **30** and the word lines **40**, and may be in contact with the active patterns **30** and the insulating capping patterns **45**, between the active patterns **30** and the insulating capping patterns **45**. For example, the gate dielectric layer **35** disposed between the first active pattern **30_1** and the first word line **40_1** may contact the first active pattern **30_1** and the first word line **40_1**.

**[0035]** The semiconductor device **1** may further include conductive pad patterns **60** that vertically overlap the active patterns **30** and are connected to the active patterns **30**.

**[0036]** Each of the conductive pad patterns **60** may include at least one conductive material layer. For example, each of the conductive pad patterns **60** may include doped silicon, metal, conductive metal nitride, metal-semiconductor compound, conductive metal oxide, conductive graphene, carbon nanotube, or combinations thereof. For example, each of the conductive pad patterns **60** may be doped silicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NON, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, or combinations thereof, but is not limited thereto. Each of the conductive pad patterns **60** may include a single layer or multiple layers of the above-described materials.

**[0037]** The conductive pad patterns **60** may be placed at a higher level than level of the word lines **40**. Each of the conductive pad patterns **60** may be connected to the upper region **30V1_U** of the first vertical active portion **30V1** and the upper region **30V2**_U of the second vertical active portion **30V2**. Each of the conductive pad patterns **60** may include an upper pad portion **60**U, a first lower pad portion **60**L**1** extending downwardly from the first side of the upper pad portion **60**U and connected to the upper region **30V1**_U of the first vertical active portion **30V1**, and a second lower pad portion **60**L**2** extending downwardly from the second side of the upper pad portion **60**U and connected to the upper region **30V2**_U of the second vertical active portion **30V2**. The width of each of the conductive pad patterns **60** in the first horizontal direction Y may be larger than the width of each of the word lines **40** in the first horizontal direction Y.

**[0038]** The conductive pad patterns **60** may include a first conductive pad pattern **60_1** and a second conductive pad pattern **60_2** adjacent to each other in the first horizontal direction Y. The first conductive pad pattern **60_1** may be connected to the upper regions **30V1**_U and **30V2**_U of the first and second vertical active portions **30V1** and **30V2** of the first active pattern **30_1**. The second conductive pad pattern **60_2** may be connected to the upper regions **30V1**_U and **30V2**_U of the first and second vertical active portions **30V1** and **30V2** of the second active pattern **30_2**.

**[0039]** In each of the active patterns **30**, the first vertical active portion **30V1** may have a first side surface **30V1_S1** facing the second vertical active portion **30V2** and a second side surface **30V1_S2** facing the first side surface **30V1_S1**, and the second vertical active portion **30V2** may have a third side surface **30V2_S1** facing the first vertical active portion **30V1** and a fourth side surface **30V2_S2** facing the third side surface **30V2_S1**.

**[0040]** In each of the conductive pad patterns **60**, the first lower pad portion **60**L**1** may contact the second side surface **30V1_S2** of the first vertical active portion **30V1**, and the second lower pad portion **60**L**2** may contact the fourth side surface **30V2_S2** of the second vertical active portion **30V2**. For example, the first lower pad portion **60**L**1** of the first conductive pad pattern **60_1** may be in contact with the second side surface **30V1_S2** of the first vertical active portion **30V1** of the first active pattern **30_1**, and the second lower pad portion **60L2** of the first conductive pad pattern **60_1** may contact the fourth side surface **30V2_S2** of the second vertical active portion **30V2** of the first active pattern **30_1**.

**[0041]** Each of the conductive pad patterns **60** may contact at least a portion of upper surfaces of the first and second vertical active portions **30V1** and **30V2**. In the first active pattern **30_1** and the first conductive pad pattern **60_1** connected to each other, a least a portion of the upper surface (**30V1**_Ta, **30V1**_Tb) of the first vertical active portion **30V1** may be in contact with the first lower pad portion **60L1**, and a least a portion of the upper surfaces **30V2**_Ta and **30V2**_Tb of the second vertical active portion **30V2** may contact the second lower pad portion **60L2**. For example, in the first active pattern **30_1** and the first conductive pad pattern **60_1** connected to each other, the upper surface (**30V1**_Ta, **30V1**_Tb) of the first vertical active portion **30V1** may include a first surface **30V1**_Ta in contact with the gate dielectric layer **35** without contacting the first lower pad portion **60L1**, and a second surface **30V1**_Tb in contact with the first lower pad portion **60L1**. The upper surface (**30V2**_Ta, **30V2**_Tb) of the second vertical active portion **30V2** may include a third surface **30V2**_Ta in contact with the gate dielectric layer **35** without contacting the second lower pad portion **60L2**, and a fourth surface **30V2**_Tb in contact with the second lower pad portion **60L2**.

**[0042]** The first lower pad portion **60L1** may be in contact with the second side surface **30V1_S2** of the first vertical active portion **30V1** and the second surface **30V1**_Tb of the upper surface (**30V1**_Ta, **30V1**_Tb) of the first vertical active portion **30V1**. The second lower pad portion **60L2** may be in contact with the fourth side surface **30V2_S2** of the second vertical active portion **30V2** and the fourth surface **30V2**_Tb of the upper surface (**30V2**_Ta, **30V2**_Tb) of the second vertical active portion **30V2**.

**[0043]** The upper capping portion **45U** may overlap the first surface **30V1**_Ta of the upper surfaces **30V1**_Ta and **30V1**_Tb of the first vertical active portion **30V1** in the vertical direction Z, and may overlap the third surface **30V2**_Ta of the upper surface **30V2**_Ta, **30V2**_Tb of the second vertical active portion **30V2** in the vertical direction Z.

**[0044]** The semiconductor device **1** may further include an insulating structure **70** disposed between the conductive pad patterns **60** and on upper surfaces of the conductive pad patterns **60**. The insulating structure **70** may include an insulating separation portion **70*a*** and an insulating capping portion **70*b***. The insulating separation portion **70*a*** is disposed between the conductive pad patterns **60** arranged in the first horizontal direction Y and extends into the insulating patterns **25**, and may be disposed between the conductive pad patterns **60** arranged in the second horizontal direction X and between the insulating capping patterns **45**. The insulating capping portion **70*b*** is disposed at a higher level than level of the conductive pad patterns **60** and may be disposed on upper surfaces of the conductive pad patterns **60**. The conductive pad patterns **60** may be spaced apart from each other by the insulating separation portion **70*a***. In the cross-sectional structure in the first horizontal direction Y as illustrated in FIG. **2**A, the lower surface of the insulating separation portion **70*a*** may contact the insulating patterns **25**, and in the cross-sectional structure in the second horizontal direction X as illustrated in FIG. **2**B, the lower surface of the insulating separation portion **70*a*** may contact the word lines **40**.

**[0045]** The semiconductor device **1** may further include an information storage structure **80**. The information storage structure **80** may be a cell capacitor capable of storing information in a memory device such as DRAM. For example, the information storage structure **80** may include first electrodes **83** connected to the conductive pad patterns **60** while penetrating through the insulating capping portion **70*b***, and extending upwardly, a dielectric layer **86** disposed on the first electrodes **83** and the insulating capping portion **70*b***, and a second electrode **89** on the dielectric layer **86**. The second electrode **89** may include a portion disposed between the first electrodes **83** and a portion disposed on upper surfaces of the first electrodes **83**.

**[0046]** In example embodiments, the word lines **40**, the active patterns **30**, and the gate dielectric layers **35** may configure cell transistors cTR. The first and second vertical active portions **30V1** and **30V2** may be vertical channel regions of the cell transistors cTR. Therefore, the integration degree of the semiconductor device **1** may be increased.

**[0047]** In example embodiments, the first word line **40_1**, the first active pattern **30_1**, and the gate dielectric layer **35** between the first word line **40_1** and the first active pattern **30_1** may configure one cell transistor cTR.

**[0048]** Since the one cell transistor cTR may use the first and second vertical active portions **30V1** and **30V2** of the first active pattern **30_1** disposed on both sides of one first word line **40_1**, for example, the first gate electrode **40_1**, as vertical channel regions, Ion (ON-Current) of the one cell transistor cTR may be improved and channel resistance may be reduced. Accordingly, the performance of the semiconductor device **1** including the cell transistor cTR may be improved.

**[0049]** In example embodiments, the first and second vertical active portions **30V1** and **30V2** may be referred to as vertical channel portions or vertical channel regions.

**[0050]** The first conductive pad pattern **60_1** may contact upper regions **30V1**_U and **30V2**_U of the first and second vertical active portions **30V1** and **30V2** of the first active pattern **30_1**, which may be vertical channel regions. Accordingly, contact resistance between the first conductive pad pattern **60_1** and the first active pattern **30_1** may be improved. Therefore, the performance of the semiconductor device **1** may be improved and reliability may be improved.

**[0051]** Hereinafter, various modifications to the above-described embodiment will be described to increase the degree of integration, improve performance, and improve reliability of the semiconductor device **1**. Various examples of modifications to the elements of the semiconductor device **1** described below will be explained with a focus on modified elements or replaced elements. In addition, the modified or replaceable elements described below are described with reference to each drawing, but the modified and replaceable elements may be combined with each other, or may be combined with other elements described above, thereby forming the semiconductor device **1** according to example embodiments.

**[0052]** A modified example of a semiconductor device according to example embodiments will be described with reference to FIG. **3**. FIG. **3** is a cross-sectional view illustrating a region taken along line I-I' of FIG. **1** to describe a modified example of a semiconductor device according to example embodiments.

**[0053]** In a variant example, referring to FIG. **3**, the first vertical active portion (**30**V**1** in FIG. **2**A) described above may be transformed into a first vertical active portion **30**V**1***a* having the upper surfaces **30**V**1**_Ta and **30**V**1**_R as in FIG. **3**, and the second vertical active portion (**30**V**2** in FIG. **2**A) having the upper surface (**30**V**2**_Ta, **30**V**2**_Tb in FIG. **2**A) described above may be transformed into a second vertical active portion **30**V**2***a* with the upper surfaces **30**V**2**_Ta and **30**V**2**_R as in FIG. **3**.

**[0054]** The upper surface (**30**V**1**_Ta, **30**V**1**_R) of the first vertical active portion (**30**V**1***a*) may include a first surface **30**V**1**_Ta in contact with the gate dielectric layer **35** without contacting the conductive pad pattern **60**, and a second surface **30**V**1**_R that is recessed beyond the first surface **30**V**1**_Ta and is in contact with the conductive pad pattern **60**. The recessed second surface **30**V**1**_R of the upper surface **30**V**1**_Ta, **30**V**1**_R of the first vertical active portion **30**V**1***a* may be in contact with the first lower pad portion **60**L**1** of the conductive pad pattern **60**.

**[0055]** The upper surface (**30**V**2**_Ta, **30**V**2**_R) of the second vertical active portion (**30**V**2***a*) may include a third surface **30**V**2**_Ta in contact with the gate dielectric layer **35** without contacting the conductive pad pattern **60**, and a fourth surface **30**V**2**_R that is recessed beyond the third surface **30**V**2**_Ta and is in contact with the conductive pad pattern **60**. The recessed fourth surface **30**V**2**_R of the upper surface **30**V**2**_Ta, **30**V**2**_R of the second vertical active portion **30**V**2***a* may be in contact with the second lower pad portion **60**L**2** of the conductive pad pattern **60**.

**[0056]** A modified example of a semiconductor device according to example embodiments will be described with reference to FIG. **4**. FIG. **4** is a cross-sectional view illustrating a region taken along line I-I' of FIG. **1** to describe a modified example of a semiconductor device according to example embodiments.

**[0057]** In a modified example, referring to FIG. **4**, the insulating capping pattern **45** including the upper capping portion **45**U described above may be transformed into an insulating capping pattern **45***a* including an upper capping portion **45**Ua as illustrated in FIG. **5**. The first side **45**S**1** of the upper capping portion **45**Ua of the insulating capping pattern **45***a* may be vertically aligned with the second side surface **30**V**1**_S**2** of the first vertical active portion **30**V**1**, and the second side **45**S**2** of the upper capping portion **45**Ua of the insulating capping pattern **45***a* may be vertically aligned with the fourth side surface **30**V**2**_S**2** of the second vertical active portion **30**V**2**.

**[0058]** The first lower pad portion **60**L**1** may contact the second side surface **30**V**1**_S**2** of the upper region **30**V**1**_U of the first vertical active portion **30**V**1**, and the upper surface **30**V**1**_T of the first vertical active portion **30**V**1** may not contact the first lower pad portion **60**L**1**. The upper surface **30**V**1**_T of the first vertical active portion **30**V**1** may contact the gate dielectric layer **35**.

**[0059]** The second lower pad portion **60**L**2** may contact the fourth side surface **30**V**4**_S**2** of the upper region **30**V**2**_U of the second vertical active portion **30**V**2**, and the upper surface **30**V**2**_T of the second vertical active portion **30**V**2** may not contact the second lower pad portion **60**L**2**. The upper surface **30**V**2**_T of the second vertical active portion **30**V**2** may contact the gate dielectric layer **35**.

**[0060]** A modified example of a semiconductor device according to example embodiments will be described with reference to FIG. **5**. FIG. **5** is a cross-sectional view illustrating a region taken along line I-I' of FIG. **1** to describe a modified example of a semiconductor device according to example embodiments.

**[0061]** In a variation, referring to FIG. **5**, the above-described active patterns (**30** in FIG. **2**A or FIG. **4**) may be modified into active patterns **30***a* further including extended active portions **30**E**1** and **30**E**2** as illustrated in FIG. **5**.

**[0062]** Each of the active patterns **30***a* may include a first extended active portion **30**E**1** extending from the upper region **30**V**1**_U of the first vertical active portion **30**V**1** in a direction away from the second vertical active portion **30**V**2**, and a second extended active portion **30**E**2** extending from the upper region **30**V**2**_U of the second vertical active portion **30**V**2** in a direction away from the first vertical active portion **30**V**1**. Therefore, each of the active patterns **30***a* may include the lower active portion **30**L, the first vertical active portion **30**V**1**, the second vertical active portion **30**V**2**, the first extended active portion **30**E**1** and the second extended active portion **30**E**2**.

**[0063]** In each of the conductive pad patterns **60**, the first lower pad portion **60**L**1** may contact the second side surface **30**V**1**_S**2** of the upper region **30**V**1**_U of the first vertical active portion **30**V**1**, and may contact the upper surface, lower surface, and side surface of the first extended active portion **30**E**1**. In each of the conductive pad patterns **60**, the second lower pad portion **60**L**2** may contact the fourth side surface **30**V**2**_S**2** of the upper region **30**V**2**_U of the second vertical active portion **30**V**2**, and may contact the upper surface, lower surface, and side surface of the second extended active portion **30**E**2**.

**[0064]** Next, with reference to FIGS. **6**, **7**A, and **7**B, a modified example of a semiconductor device according to example embodiments will be described. Referring to FIGS. **6**, **7**A, and **7**B, FIG. **6** is a plan view illustrating a semiconductor device according to example embodiments, FIG. **7**A is a cross-sectional view conceptually illustrating the area taken along line Ia-Ia' in FIG. **6**, and FIG. **7**B is a partial enlarged view of the area indicated by IIa-IIa' in FIG. **6**.

**[0065]** Referring to FIGS. **6**, **7**A, and **7**B, the semiconductor device **100** according to example embodiments may include the lower structure LS described with reference to FIGS. **1**, **2**A, and **2**B.

**[0066]** The semiconductor device **100** may further include insulating patterns **125** disposed on the lower structure LS. Each of the insulating patterns **125** may have a line shape extending in the second horizontal direction X. The insulating patterns **125** may be formed of an insulating material such as silicon oxide.

**[0067]** The semiconductor device **100** may further include active patterns **130** arranged in the first horizontal direction Y and the second horizontal direction X. The active patterns **130** may be formed of the same semiconductor material as the previously described active patterns (FIGS. **1**, **2**A, and **2**B). The active patterns **130** may include an oxide semiconductor.

**[0068]** Each of the active patterns **130** may include a lower active portion **130**L, a first vertical active portion **130**V**1**, a second vertical active portion **130**V**2**, a first extended active portion **130**E**1**, and a second extended active portion **30**E**2**. In each of the active patterns **130**, the first vertical active portion **130**V**1** and the second vertical active portion **130**V**2**

may be arranged in the first horizontal direction Y. The lower active portion **130**L may be disposed on the bit line **6** and connected to the bit line **6**. For example, the lower surface of the lower active portion **130**L may contact the upper surface of the bit line **6**. The first vertical active portion **130**V**1** may extend upwardly from the first side of the lower active portion **130**L. The first vertical active portion **130V1** may include a first lower region **130**V**1**_L and a first upper region **130**V**1**_U on the first lower region **130**V**1**_L. The second vertical active portion **130V2** may extend upwardly from the second side of the lower active portion **130**L. The second vertical active portion **130V2** may include a first lower region **130**V**2**_L and a first upper region **130**V**2**_U on the first lower region **130**V**2**_L.

**[0069]** The first extended active portion **130**E**1** may extend from the upper region **130**V**1**_U of the first vertical active portion **130V1** in a direction away from the second vertical active portion **130V2**. The first extended active portion **130**E**1** may extend in a direction away from the second vertical active portion **130V2** from the upper end of the first vertical active portion **130V1**. A lower surface of the first extended active portion **130E1** may contact an upper surface of the insulating pattern **125**. The second extended active portion **130**E**2** may extend from the upper region **130**V**2**_U of the second vertical active portion **130V2** in a direction away from the first vertical active portion **130V1**. The second extended active portion **130E2** may extend from the upper end of the second vertical active portion **230V1** in a direction away from the first vertical active portion **130V1**. A lower surface of the second extended active portion **130E2** may contact an upper surface of the insulating pattern **125**. The upper surfaces **130**V**1**_T and **130**V**2**_T of the first and second extended active portions (**130**E**1** and **30**E**2** in FIG. **7**A) may be flat.

**[0070]** The semiconductor device **100** may further include word lines **140**. The word lines **140** may be gate electrodes. The word lines **140** may be the same as the word lines (**40** in FIGS. **1**, **2**A, and **2**B) described above.

**[0071]** The word lines **140** may be disposed on a lower level than level of the upper surfaces of the first and second vertical active portions **130V1** and **130V2**. The word lines **140** may be placed at a lower level than level of the upper surfaces of the insulating patterns **125**. Each of the word lines **140** may have a line shape extending in the second horizontal direction X. The word lines **140** may include a first word line **140_1** and a second word line **140_2** that are adjacent to each other in the first horizontal direction Y. The word lines **140** may be disposed on the lower active portions **130**L and between the first vertical active portions **130V1** and the second vertical active portions **130V2**.

**[0072]** The semiconductor device **100** may further include insulating capping patterns **145** on the word lines **140**. The insulating capping patterns **145** may be disposed between the first vertical active portions **130V1** and the second vertical active portions **130V2**. The insulating capping patterns **145** may contact upper surfaces of the word lines **140**. The insulating capping patterns **145** may be disposed at a lower level than level of the first and second extended active portions **130E1** and **130E2**.

**[0073]** The semiconductor device **100** may further include gate dielectric layers **35** disposed between the active patterns **130** and the word lines **140** and between the active patterns **130** and the insulating capping patterns **145**. The gate dielectric layers **35** may contact the active patterns **30**, the word lines **40**, and the insulating capping patterns **45**.

**[0074]** The semiconductor device **100** may further include a dummy insulating structure **148** disposed on the first and second extended active portions **130**E**1** and **130**E**2** of the active patterns **130**. The dummy insulating structure **148** may cover a portion of the upper surfaces **130**V**1**_T and **130**V**2**_T of the first and second extended active portions **130**E**1** and **130**E**2**. The dummy insulating structure **148** may include a dummy dielectric layer **137** and a dummy capping insulating pattern **147** that are sequentially stacked. The dummy dielectric layer **137** may be formed of the same material as the gate dielectric layer **135** and may have the same thickness as the gate dielectric layer **135**. The dummy capping insulating pattern **147** may be formed of the same material as the insulating capping patterns **145**.

**[0075]** The semiconductor device **100** may further include conductive pad patterns **160** that vertically overlap the word lines **140** and are connected to the active patterns **130**. The conductive pad patterns **160** may be formed of the same material as the conductive pad patterns (**60** in FIGS. **1**, **2**A, and **2**B) described above. The conductive pad patterns **160** may be placed at a higher level than level of the word lines **140**. Each of the conductive pad patterns **160** may be connected to the upper region **130**V**1**_U of the first vertical active portion **130V1** and the upper region **130**V**2**_U of the second vertical active portion **130V2**. Each of the conductive pad patterns **160** may include an upper pad portion **160**U, and a lower pad portion **160**L extending downwardly from the center area of the upper pad portion **160**U and connected to the upper region **130**V**1**_U of the first vertical active portion **130V1** and the upper region **130**V**2**_U of the second vertical active portion **130V2**. The width of each of the conductive pad patterns **160** in the first horizontal direction Y may be greater than the width of each of the word lines **140** in the first horizontal direction Y.

**[0076]** The conductive pad patterns **160** may include a first conductive pad pattern **160_1** and a second conductive pad pattern **160_2** adjacent to each other in the first horizontal direction Y. The active patterns **130** may include a first active pattern **130_1** and a second active pattern **130_2** adjacent to each other in the first horizontal direction Y.

**[0077]** The first conductive pad pattern **160_1** may be connected to the upper regions **130V1_**U and **130V2_**U of the first and second vertical active portions **130V1** and **130V2** of the first active pattern **130_1**. The second conductive pad pattern **160_2** may be connected to the upper regions **130V1_**U and **130V2_**U of the first and second vertical active portions **130V1** and **130V2** of the second active pattern **130_2**.

**[0078]** In each of the active patterns **130**, the first vertical active portion **130V1** may have a first side surface **130**V**1**_S**1** facing the second vertical active portion **130V2** and a second side surface **130V1_S2** facing the first side **130**V**1**_S**1**, and the second vertical active portion **130V2** may have a third side surface **130V2_S1** facing the first vertical active portion **130V1** and a fourth side surface **130V2_S2** facing the third side surface **130V2_S1**.

**[0079]** In each of the conductive pad patterns **160**, the lower pad portion **160**L may contact the first side **130V1_S1** of the first vertical active portion **130V1** and the third side surface **30V2_S1** of the second vertical active portion **130V2**.

**[0080]** Each of the conductive pad patterns **160** may contact at least a portion of the upper surfaces **130**V**1**_T and **103**V**2**_T of the first and second vertical active portions **130**V**1** and **130**V**2**. For example, in the first active pattern **130_1** and the first conductive pad pattern **160_1** connected to each other, the first conductive pad pattern **160_1** may be in contact with at least a portion of the upper surface **130**V**1**_T of the first vertical active portion **130**V**1** and at least a portion of the upper surface **130**V**2**_T of the second vertical active portion **130**V**2**.

**[0081]** The upper surface **130**V**1**_T of the first extended active portion **130**E**1** may include a portion in contact with the dummy insulating structure **148** and a portion in contact with the conductive pad pattern **160**. The upper surface **130**V**2**_T of the second extended active portion **130**E**2** may include a portion in contact with the dummy insulating structure **148** and a portion in contact with the conductive pad pattern **160**.

**[0082]** The semiconductor device **100** may further include an insulating structure **170** disposed on the upper surfaces of the conductive pad patterns **160**, disposed between the conductive pad patterns **160**, and extending downwardly to pass between the active patterns **130**.

**[0083]** The insulating structure **170** may include an insulating separation portion **170***a* and an insulating capping portion **170***b*.

**[0084]** The insulating separation portion **170***a* may be disposed between the conductive pad patterns **60** arranged in the first horizontal direction Y, penetrate the dummy insulating structure **148**, extend downwardly, pass between the active patterns **130** to extend into the insulating patterns **125**, and may be disposed between the conductive pad patterns **160** arranged in the second horizontal direction X and between the insulating capping patterns **145**.

**[0085]** The insulating capping portion **170***b* is disposed at a higher level than level of the conductive pad patterns **160** and may be disposed on upper surfaces of the conductive pad patterns **160**. The conductive pad patterns **160** may be spaced apart from each other by the insulating separation portion **170***a*. In the cross-sectional structure in the first horizontal direction Y as illustrated in FIG. **7**A, the lower surface of the insulating separation portion **170***a* may contact the insulating patterns **125**, and in the cross-sectional structure in the second horizontal direction X as illustrated in FIG. **7**B, the lower surface of the insulating separation portion **170***a* may contact the word lines **140**. The insulating separation portion **170***a* may contact side surfaces of the extended active portions **130**E**1** and **130**E**2** of the active patterns **130**. The lower surface of the insulating separation portion **170***a* may be disposed at a lower level than level of the extended active portions **130**E**1** and **130**E**2**.

**[0086]** The semiconductor device **100** may further include the same information storage structure **80** as described above. The information storage structure **80** may be a cell capacitor capable of storing information in a memory device such as DRAM. For example, the information storage structure **80** may include first electrodes **83** penetrating the insulating capping portion **170***b*, connected to the conductive pad patterns **160** and extending upwardly, a dielectric layer **86** disposed on the first electrodes **83** and the insulating capping portion **170***b*, and a second electrode **89** on the dielectric layer **86**.

**[0087]** A modified example of a semiconductor device according to example embodiments will be described with reference to FIG. **8**. FIG. **8** is a cross-sectional view illustrating a region taken along line Ia-Ia' of FIG. **6** to describe a modified example of a semiconductor device according to example embodiments.

**[0088]** In a variation, referring to FIG. **8**, the first and second extended active portions (**130**E**1** and **30**E**2** in FIG. **7**A) described above may be modified into the first and second extended active portions **130**E**1***a* and **130**E**2***a* as illustrated in FIG. **8**. The upper surfaces **130**V**1**_Ta and **130**V**1**_Tb of the first extended active portion **130**E**1***a* may include a recessed first surface **130**V**1**_Ta and a flat second surface **130**V**1**_Tb, and the upper surfaces **130**V**2**_Ta and **130**V**2**_Tb of the second extended active portion **130**E**2***a* may include a recessed third surface **130**V**2**_Ta and a flat fourth surface **130**V**2**_Tb. In the case of the upper surfaces **130**V**1**_Ta and **130**V**1**_Tb of the first extended active portion **130**E**1***a*, the recessed first surface **130**V**1**_Ta may be recessed than the flat second surface **130**V**1**_Tb, and in the case of the upper surfaces **130**V**2**_Ta and **130**V**2**_Tb of the second extended active portion **130**E**2***a*, the recessed third surface **130**V**2**_Ta may be recessed more than the flat fourth surface **130**V**2**_Tb. The recessed first surface **130**V**1**_Ta of the upper surface **130**V**1**_Ta, **130**V**1**_Tb of the first extended active portion **130**E**1***a*, and the recessed third surface **130**V**2**_Ta of the upper surface **130**V**2**_Ta, **130**V**2**_Tb of the second extended active portion **130**E**2***a*, may contact the conductive pad pattern **160**.

**[0089]** A modified example of a semiconductor device according to example embodiments will be described with reference to FIG. **9**. FIG. **9** is a cross-sectional view illustrating a region taken along line Ia-Ia' of FIG. **6** to describe a modified example of a semiconductor device according to example embodiments.

**[0090]** In a modified example, referring to FIG. **9**, the previously described dummy insulating structure **148** in FIG. **7**A may be transformed into the dummy insulating structure **148***a* as illustrated in FIG. **9**. The dummy insulating structure **148***a* may cover or overlap the entire upper surface **130**V**1**_T and **130**V**2**_T of the active pattern **130**. The dummy insulating structure **148***a* may include a dummy dielectric layer **137***a* and a dummy capping insulating layer **147***a* corresponding to the previously described dummy dielectric layer (**137** in FIG. **7**A) and the dummy capping insulating layer (**147** in FIG. **7**A), respectively. The lower pad portion **160**L**1** of the conductive pad pattern **160** may contact the first side **130**V**1**_S**1** of the first vertical active portion **130**V**1** and the third side surface **30**V**2**_S**1** of the second vertical active portion **130**V**2**. The upper surfaces (**130**V**1**_T, **130**V**2**_T) of the active pattern **130** may not be in contact with the conductive pad pattern **160**.

**[0091]** A modified example of a semiconductor device according to example embodiments will be described with reference to FIG. **10**. FIG. **10** is a cross-sectional view illustrating a region taken along line Ia-Ia' of FIG. **6** to describe a modified example of a semiconductor device according to example embodiments.

**[0092]** In a modified example, referring to FIG. **10**, the dummy insulating structure (**148** in FIG. **7**A) described above may be omitted, and the conductive pad pattern (**160** in FIG. **7**A) described above may be transformed into a conductive pad pattern **160***c* that contacts the first and second extended active portions **130**E**1** and **130**E**2** of the active pattern **130**, as illustrated in FIG. **10**.

**[0093]** The conductive pad pattern **160***c* may include an upper pad portion **160**Uc and a lower pad portion **160**Lc extending downwardly from the center area of the upper pad portion **160**Uc. The upper pad portion **160**Uc may contact the first and second extended active portions **130**E**1** and **130**E**2** of the active pattern **130**, and the lower pad portion **160**Lc may contact the first side **130**V**1**_S**1** of the first vertical active portion **130**V**1** and the third side surface **30**V**2**_S**1** of the second vertical active portion **130**V**2**.

**[0094]** Next, an example of a method of forming a semiconductor device according to example embodiments will be described with reference to FIG. **1** and FIGS. **11**A, **11**B, **12**, **13**A, **13**B, **14**A, and **14**B. In FIGS. **11**A, **11**B, **12**, **13**A, **13**B, **14**A, and **14**B, FIGS. **11**A, **12**, **13**A, and **14**A are cross-sectional views illustrating the area taken along line I-I' of FIG. **1**, and FIGS. **11**B, **13**B, and **14**B are cross-sectional views illustrating the area taken along line II-II' in FIG. **1**.

**[0095]** Referring to FIGS. **1**, **11**A, and **11**B, the lower structure LS may be formed. Forming the lower structure LS may include forming bit lines **6** on the base **3** and forming a shield structure **9** between the bit lines **6**. The bit lines **6** may be conductive lines. Each of the bit lines **6** may have a line shape extending in the first horizontal direction Y. Forming the shield structure **9** may include forming a shield insulating layer **12** covering the inner wall of the space between the conductive lines **6**, and forming a shield conductive line **15** and a shield insulating capping pattern **18** sequentially laminated on the shield insulating layer **12**.

**[0096]** Preliminary insulating patterns **24** may be formed on the lower structure LS. Each of the preliminary insulating patterns **24** may have a line shape extending in a second horizontal direction X perpendicular to the first horizontal direction Y.

**[0097]** A material layer covering, overlapping, or on the lower structure LS and the preliminary insulating patterns **24** may be formed, and the material layer may be patterned to form material patterns **29**. Each of the material patterns **29** extends in the first horizontal direction Y and may cover the lower structure LS and the preliminary insulating patterns **24**. The material patterns **29** may cover, overlap, or be on the upper and side surfaces of the preliminary insulating patterns **24** and the lower structure LS between the preliminary insulating patterns **24**. The material patterns **29** may be spaced apart from each other in the second horizontal direction X. The material patterns **29** may be formed of a semiconductor material such as an oxide semiconductor.

**[0098]** A gate dielectric layer **35** may be formed to conformally cover, overlap, or be on the lower structure LS, the preliminary insulating patterns **24**, and the material patterns **29**.

**[0099]** On the gate dielectric layer **35**, word lines **40** may be formed to partially fill the space between the preliminary insulating patterns **24**. The word lines **40** may be gate electrodes.

**[0100]** A capping layer **43** may be formed on the gate dielectric layer **35** and the word lines **40**. First mask patterns **50** having openings **50***a* may be formed on the capping layer **43**. The first mask patterns **50** may vertically overlap the word lines **40** and may be formed to have a width larger than the word lines **40**. The openings **50***a* may vertically overlap the preliminary insulating patterns **24**.

**[0101]** Referring to FIGS. **1** and **12**, as an etching process is performed using the first mask patterns **50** as an etch mask, the capping layer **43**, the gate dielectric layer **35**, and the material patterns **29** may be sequentially etched, and then the preliminary insulating patterns **24** may be partially etched. The capping layer **43** may be etched to form capping lines **44**, the material patterns **29** are etched to form active patterns **30**, and the preliminary insulating patterns **24** may be partially etched to form insulating patterns **25** with a reduced height.

**[0102]** Each of the capping lines **44** may have a line shape extending in the second horizontal direction X. The capping lines **44** may be formed on the word lines **40**. Each of the active patterns **30** may include a lower active portion **30**L connected to the upper surface of the bit line **6**, a first vertical active portion **30**V**1** extending upwardly from the first side of the lower active portion **30**L, and a second vertical active portion **30**V**2** extending upwardly from the second side of the lower active portion **30**L. The insulating patterns **25** may be disposed at a higher level than level of the upper surfaces of the word lines **40** and may have upper surfaces disposed at a lower level than the level of upper ends of the active patterns **30**. Accordingly, upper regions of the first and second vertical active portions **30**V**1** and **30**V**2** of the active patterns **30** may be exposed by the insulating patterns **25**. The first mask patterns **50** may be removed.

**[0103]** Referring to FIGS. **1**, **13**A, and **13**B, a conductive pad layer **59** may be formed on the insulating patterns **25**, the active patterns **30**, and the capping lines **44**. The conductive pad layer **59** may contact upper regions of the first and second vertical active portions **30**V**1** and **30**V**2** of the active patterns **30** exposed by the insulating patterns **25**.

**[0104]** Referring to FIGS. **1**, **14**A, and **14**B, a second mask pattern **65** may be formed on the conductive pad layer **59**. An etching process using the second mask pattern **65** as an etch mask may be performed to etch the conductive pad layer **59** and the capping lines **44**. The conductive pad layer **59** may be etched to form conductive pad patterns **60**, and the capping lines **44** may be etched to form insulating capping patterns **45**.

**[0105]** Each of the conductive pad patterns **60** may include an upper pad portion **60**U, a first lower pad portion **60**L**1** extending downwardly from the first side of the upper pad portion **60**U and connected to an upper region of the first vertical active portion **30**V**1**, and a second lower pad portion **60**L**2** extending downwardly from the second side of the upper pad portion **60**U and connected to the upper region of the second vertical active portion **30**V**2**. Each of the insulating capping patterns **45** may include an upper capping portion disposed at a higher level than level of the active pattern **30** and a lower capping portion extending downwardly from the center of the upper capping portion and contacting the upper surface of the word line **40**. The insulating capping pattern **45** may include at least one vertically overlapping portion of the first and second vertical active portions **30**V**1** and **30**V**2**.

**[0106]** Referring again to FIGS. **1**, **2**A, and **2**B, after removing the second mask pattern **65**, the insulating structure **70** may be formed. The insulating structure **70** may include an insulating separation portion **70***a* and an insulating capping portion **70***b*. The insulating separation portion **70***a* may extend into the insulating patterns **25** while partially or completely filling a gap between the conductive pad patterns **60** arranged in the first horizontal direction Y, and may partially or completely fill a gap between the conductive pad patterns **60** and the insulating capping patterns **45** arranged in the second horizontal direction X. The insulating capping portion **70***b* may be disposed at a higher level than level of the conductive pad patterns **60** and may be formed on upper surfaces of the conductive pad patterns **60**.

**[0107]** An information storage structure **80** may be formed. The information storage structure **80** may be a cell capacitor capable of storing information in a memory device such as DRAM. For example, the information storage structure **80** may include first electrodes **83** penetrating the insulating capping portion **70***b*, connected to the conductive pad patterns **60** and extending upwardly, a dielectric layer **86** disposed on the first electrodes **83** and the insulating capping portion **70***b*, and a second electrode **89** on the dielectric layer **86**.

**[0108]** Next, another example of a method of forming a semiconductor device according to example embodiments will be described with reference to FIG. **6** and FIGS. **15**A, **15**B, **16**A, **16**B, **17**A, **17**B, **18**A, and **18**B. In FIGS. **15**A, **15**B, **16**A, **16**B, **17**A, **17**B, **18**A, and **18**B, FIGS. **15**A, **16**A, **17**A, and **18**A are cross-sectional views illustrating the area taken along line Ia-Ia' of FIG. **6**, and FIGS. **15***b*, **16***b*, **17***b*, and **18***b* are cross-sectional views illustrating the area taken along line IIa-IIa' in FIG. **6**.

**[0109]** Referring to FIGS. **6**, **15**A, and **15**B, the lower structure LS as illustrated in FIGS. **11**A and **11**B may be formed. Insulating patterns **125** may be formed on the lower structure LS. Each of the insulating patterns **125** may have a line shape extending in a second horizontal direction X perpendicular to the first horizontal direction Y.

**[0110]** A material layer covering, overlapping, of on the lower structure LS and the insulating patterns **125** may be formed, and the material layer may be patterned to form material patterns **129**. The material patterns **129** may be the same as the material patterns **29** described in FIGS. **11**A and **11**B.

**[0111]** A gate dielectric layer **135** may be formed to conformally cover or overlap the lower structure LS, the insulating patterns **125**, and the material patterns **129**. On the gate dielectric layer **135**, word lines **140** may be formed to partially fill the space between the insulating patterns **125**. The word lines **140** may be gate electrodes.

**[0112]** A capping layer **143** may be formed on the gate dielectric layer **135** and the word lines **140**. First mask patterns **150** having openings **150***a* may be formed on the capping layer **143**. The first mask patterns **150** may vertically overlap the insulating patterns **125**. The openings **150***a* may vertically overlap the word lines **140**. The openings **150***a* may be formed to have a width larger than the width of the word lines **40**.

**[0113]** Referring to FIGS. **6**, **16**A, and **16**B, an etching process using the first mask patterns **150** as an etch mask may be performed to sequentially etch the capping layer **143** and the gate dielectric layer **135**, and side surfaces of upper regions of the material patterns **129** may be exposed.

**[0114]** The capping layer **143** may be etched to form dummy capping lines **146** and capping lines **144**. The capping lines **144** may be formed on the word lines **140** and may be formed at a lower level than level of the upper surfaces of the material patterns **129**. The dummy capping lines **146** may vertically overlap upper surfaces of the insulating patterns **125**. The dummy capping lines **146** may be placed at a higher level than level of the capping lines **144**. Among the gate dielectric layers **135**, the gate dielectric layers located in areas that overlap perpendicularly to the upper surfaces of the insulating patterns **125** may be formed into dummy dielectric lines **136** by the etching process. Lower surfaces of the dummy capping lines **146** may contact upper surfaces of the dummy dielectric lines **136**.

**[0115]** Referring to FIGS. **6**, **17**A, and **17**B, after removing the first mask patterns **150**, a conductive pad layer **159** may be formed. The conductive pad layer **159** may cover, overlap, or be on the dummy capping lines **146** and the capping lines **144** and may contact side surfaces of upper regions of the material patterns **129**.

**[0116]** Referring to FIGS. **6**, **18**A, and **18**B, a second mask pattern **165** may be formed on the conductive pad layer **159**. As an etching process is performed using the second mask pattern **165** as an etch mask, the conductive pad layer **159**, the dummy capping lines **46**, the dummy dielectric lines **136**, the material patterns **129**, and the capping lines **144** may be etched.

**[0117]** The conductive pad layer **159** may be etched to form conductive pad patterns **160**, the dummy capping lines **146** may be etched to form dummy capping insulating patterns **147**, the dummy dielectric lines **136** may be etched to form dummy dielectric layers **137**, the material patterns **129** may be etched to form active patterns **130** spaced apart from each other, and the capping lines **144** may be etched to form insulating capping patterns **145**. The dummy insulating structure **148** may include the dummy dielectric layer **137** and the dummy capping insulating pattern **147** that are sequentially stacked.

**[0118]** Each of the active patterns **130** may include a lower active portion **130**L, a first vertical active portion **130V1**, a second vertical active portion **130**V**2**, a first extended active portion **130**E**1**, and a second extended active portion **30**E**2**. In each of the active patterns **130**, the lower active portion **130**L may be formed on the bit line **6**, the first vertical active portion **130V1** may extend upwardly from the first side of the lower active portion **130**L, the second vertical active portion **130**V**2** may extend upwardly from the second side of the lower active portion **130**L, the first extended active portion **130E1** may extend from an upper region of the first vertical active portion **130V1** in a direction away from the second vertical active portion **130V2**, and the second extended active portion **130**E**2** may extend from an upper region of the second vertical active portion **130V2** in a direction away from the first vertical active portion **130V1**. Lower surfaces of the first and second extended active portions **130**E**1** and **130**E**2** may contact upper surfaces of the insulating patterns **125**.

**[0119]** Each of the conductive pad patterns **160** may be connected to an upper region of the first vertical active portion **130V1** and an upper region of the second vertical active portion **130**V**2**. Each of the conductive pad patterns **160** may include an upper pad portion **160**U, and a lower pad portion **160**L extending downwardly from the middle region of the upper pad portion **160**U and connected to the upper region of the first vertical active portion **130V1** and the upper region of the second vertical active portion **130V2**. For example, the lower pad portion **160**L may contact a side surface of the upper region of the first vertical active portion **130V1** and a side surface of the upper region of the second vertical active portion **130**V**2**.

**[0120]** Referring again to FIGS. **1**, **7**A, and **7**B, after removing the second mask pattern **165**, the insulating structure **170** may be formed. The insulating structure **170** may include an insulating separation portion **170***a* and an insulating capping portion **170***b*.

**[0121]** The insulating structure **170** may include an insulating separation portion **170***a* and an insulating capping portion **170***b*. The insulating separation portion **170***a* may partially or completely fill space between the conductive pad patterns **60** arranged in the first horizontal direction Y, extend downwardly, pass between the active patterns **130** while extending into the insulating patterns **125**, and fill space between the conductive pad patterns **160** arranged in the second horizontal direction X and between the insulating capping patterns **145**. The insulating capping portion **170***b* is disposed at a higher level than level of the conductive pad patterns **160** and may be disposed on upper surfaces of the conductive pad patterns **160**. The conductive pad patterns **160** may be spaced apart from each other by the insulating separation portion **170***a*. The insulating separation portion **170***a* may contact side surfaces of the extended active portions **130**E**1** and **130**E**2** of the active patterns **130**. The lower surface of the insulating separation portion **170***a* may be formed at a lower level than level of the extended active portions **130**E**1** and **130**E**2**.

**[0122]** An information storage structure **180** may be formed. The information storage structure **180** may be a cell capacitor capable of storing information in a memory device such as DRAM. For example, the information storage structure **180** may include first electrodes **83** penetrating through the insulating capping portion **170***b*, connected to the conductive pad patterns **160** and extending upwardly, a dielectric layer **86** disposed on the first electrodes **83** and the insulating capping portion **70***b*, and a second electrode **89** on the dielectric layer **86**.

**[0123]** As set forth above, according to example embodiments, a semiconductor device including an active pattern containing first and second vertical active portions on both sides of one gate electrode, and a conductive pad pattern connected to upper regions of the first and second vertical active portions may be provided. The first and second vertical active portions may be vertical channel regions of one transistor. Since the conductive pad pattern may contact upper regions of the first and second vertical active portions of the active pattern, contact resistance between the conductive pad pattern and the active pattern may be improved. Therefore, the Ion (ON-Current) of the transistor may be improved and the channel resistance and contact resistance may be reduced, thereby improving the performance of the semiconductor device.

**[0124]** While example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concept as defined by the appended claims.

**1**. A semiconductor device comprising:

a bit line;

an active pattern on the bit line, and including a lower active portion electrically connected to the bit line, a first vertical active portion extending from the lower active portion, and a second vertical active portion extending from the lower active portion, wherein the first vertical active portion and the second vertical active portion are spaced apart from one another;

a word line between a lower region of the first vertical active portion and a lower region of the second vertical active portion; and

a conductive pad pattern electrically connected to an upper region of the first vertical active portion and an upper region of the second vertical active portion.

**2**. The semiconductor device of claim **1**, wherein the active pattern includes an oxide semiconductor.

**3**. The semiconductor device of claim **1**, wherein the bit line extends in a first horizontal direction,

wherein the word line extends in a second horizontal direction perpendicular to the first horizontal direction,

wherein the first vertical active portion and the second vertical active portion are spaced apart in the first horizontal direction, and

wherein the conductive pad pattern comprises:

an upper pad portion;

a first lower pad portion extending from the upper pad portion and electrically connected to the upper region of the first vertical active portion; and

a second lower pad portion extending from the upper pad portion and electrically connected to the upper region of the second vertical active portion.

**4**. The semiconductor device of claim **3**, wherein the first vertical active portion has a first side surface facing the second vertical active portion and a second side surface opposing the first side surface,

wherein the second vertical active portion has a third side surface facing the first vertical active portion and a fourth side surface opposing the third side surface,

wherein the first lower pad portion contacts the second side surface of the first vertical active portion, and

wherein the second lower pad portion contacts the fourth side surface of the second vertical active portion.

**5**. The semiconductor device of claim **4**, wherein at least a portion of an upper surface of the first vertical active portion is in contact with the first lower pad portion.

**6**. The semiconductor device of claim **1**, wherein an upper surface of the first vertical active portion has a first surface not in contact with the conductive pad pattern, and a second surface further recessed than the first surface and is in contact with the conductive pad pattern.

**7**. The semiconductor device of claim **1**, further comprising an insulating capping pattern between an upper surface of the word line and the conductive pad pattern,

wherein the insulating capping pattern comprises:

an upper capping portion that is a greater first distance from the bit line than a second distance of the active pattern from the bit line; and

a lower capping portion extending from a central area of the upper capping portion and contacting the upper surface of the word line, and

wherein the upper capping portion includes a portion vertically overlapping at least one of the first and second vertical active portions.

**8**. The semiconductor device of claim **1**, wherein the active pattern further comprises:

a first extended active portion extending from an upper region of the first vertical active portion in a direction away from the second vertical active portion; and

a second extended active portion extending from an upper region of the second vertical active portion in a direction away from the first vertical active portion.

**9**. The semiconductor device of claim **8**, wherein the conductive pad pattern comprises:

an upper pad portion;

a first lower pad portion extending from the upper pad portion and contacting an upper surface, a lower surface and a side surface of the first extended active portion; and

a second lower pad portion extending from the upper pad portion and contacting an upper surface, a lower surface and a side surface of the second extended active portion.

**10**. The semiconductor device of claim **8**, wherein the first vertical active portion has a first side surface facing the second vertical active portion and a second side surface opposing the first side surface,

wherein the second vertical active portion has a third side surface facing the first vertical active portion and a fourth side surface opposing the third side surface, and

wherein the conductive pad pattern comprises:

an upper pad portion; and

a lower pad portion extending from a central area of the upper pad portion and contacting the first side surface of the first vertical active portion and the third side surface of the second vertical active portion.

**11**. The semiconductor device of claim **10**, wherein at least a portion of an upper surface of the active pattern is in contact with the conductive pad pattern.

**12**. A semiconductor device comprising:

a conductive line extending in a first horizontal direction;

a first active pattern and a second active pattern on the conductive line;

a first gate electrode and a second gate electrode on the conductive line; and

a first conductive pad pattern and a second conductive pad pattern on the first and second gate electrodes, respectively,

wherein each of the first and second active patterns comprises:

a lower active portion electrically connected to an upper surface of the conductive line;

a first vertical active portion extending from the lower active portion; and

a second vertical active portion extending from the lower active portion,

wherein the first gate electrode is between the first vertical active portion of the first active pattern and the second vertical active portion of the first active pattern, and is on the lower active portion of the first active pattern,

wherein the second gate electrode is between the first vertical active portion of the second active pattern and the second vertical active portion of the second active pattern, and is on the lower active portion of the first second active pattern,

wherein the first conductive pad pattern is electrically connected to the first vertical active portion of the first active pattern and the second vertical active portion of the first active pattern, and

wherein the second conductive pad pattern is electrically connected to the first vertical active portion of the second active pattern and the second vertical active portion of the second active pattern.

**13**. The semiconductor device of claim **12**, wherein respective widths of the first and second conductive pad patterns in the first horizontal direction are greater than respective widths of the first and second active patterns in the first horizontal direction.

**14**. The semiconductor device of claim **12**, wherein the first conductive pad pattern comprises:

an upper pad portion on the first active pattern;

a first lower pad portion extending from the upper pad portion and contacting a side surface of the first vertical active portion of the first active pattern; and

a second lower pad portion extending from the upper pad portion and contacting a side surface of the second vertical active portion of the first active pattern.

**15**. The semiconductor device of claim **14**, wherein the first lower pad portion is in contact with at least a portion of an upper surface of the first vertical active portion.

**16**. The semiconductor device of claim **12**, wherein the first conductive pad pattern comprises:

an upper pad portion on the first active pattern; and

a lower pad portion extending from a central portion of the upper pad portion and contacting a side surface of the first vertical active portion of the first active pattern and a side surface of the second vertical active portion of the first active pattern.

**17**. The semiconductor device of claim **16**, wherein the lower pad portion is in contact with at least a portion of an upper surface of the first vertical active portion.

**18**. A semiconductor device comprising:

a conductive line extending in a first horizontal direction;

a cell transistor on the conductive line; and

a conductive pad pattern on the cell transistor,

wherein the cell transistor comprises:

an active pattern including a first vertical channel portion and a second vertical channel portion spaced apart from each other in the first horizontal direction;

a single gate electrode between the first and second vertical channel portions; and

a gate dielectric layer between the single gate electrode and the active pattern, and

wherein the conductive pad pattern contacts upper regions of the first and second vertical channel portions.

**19**. The semiconductor device of claim **18**, wherein the first vertical channel portion has a first side surface facing the second vertical channel portion and a second side surface opposing the first side surface,

wherein the second vertical channel portion has a third side surface facing the first vertical channel portion and a fourth side surface opposing the third side surface, and

wherein the conductive pad pattern comprises:

an upper pad portion;

a first lower pad portion extending from the upper pad portion and contacting the second side surface of the first vertical channel portion; and

a second lower pad portion extending from the upper pad portion and contacting the fourth side surface of the second vertical channel portion.

**20**. The semiconductor device of claim **18**, wherein the first vertical channel portion has a first side surface facing the second vertical channel portion and a second side surface opposing the first side surface,

wherein the second vertical channel portion has a third side surface facing the first vertical channel portion and a fourth side surface opposing the third side surface, and

wherein the conductive pad pattern comprises:
an upper pad portion; and
a lower pad portion extending from a central area of the upper pad portion and contacting the first side surface of the first vertical channel portion and the third side surface of the second vertical channel portion.

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